NOTES FOR RENORMALIZATION FLOW

Based on the paper by A.Dunlap and Cole

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Contents

1 Setup

1.1 Checklist

1. Theorem 3.2.2. I should be some interval such that $\sigma > 0$.
   2. Finish the argument that Sym is a vector space.
   3. Well-definedness of the Wiener integral.

1.2 Semilinear SHE

We consider the semilinear stochastic heat equation

$$du_t^\rho = \frac{1}{2}\Delta u_t^\rho dt + \gamma_\rho \sigma(u_t^\rho) dW_t^\rho, \quad t > 0, x \in \mathbb{R}^2$$

Here σ is a Lipschitz nonlinearity and $dW_t^\rho(x)$ is a Gaussian noise that is white in time and correlated in space at scale $\rho^{1/2} << 1$. We are interested in the pointwise behavior of $u_t^\rho(x)$ as $\rho \to 0$, which calls for an attenuation factor $\gamma_\rho \sim |\ln \rho|^{-1/2}$ due to critical scaling in two dimensions. In fact, we devote most of our attention to a variation on (**??**) in which we first multiply σ and then smooth the noise:

$$dv_t^\rho = \frac{1}{2}\Delta v_t^\rho dt + \gamma_\rho \mathcal{G}_\rho[\sigma(v_t^\rho) dW_t]$$

Definiton 1.2.1.

(Space-time White Noise)

Let $dW = (dW_t(x))_{t\in\mathbb{R}, x\in\mathbb{R}^2}$ be a standard $\mathbb{R}^m$-valued space-time white noise generating a temporal filtration $\{\mathcal{F}_t\}_{t\in\mathbb{R}}$. Writing $dW = (dW^1, \cdots, dW^m)$ in components, then

$$\mathbb{E}[dW_t^i(x) dW_{t'}^{i'}(x')] = \delta_{i,i'}\delta(t-t')\delta(x-x')$$

Proposition 1.2.1. Construct a space-time white noise.

Definiton 1.2.2.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and fix a target dimension $m \in \mathbb{N}$. The solution $v^\rho : \Omega \times \mathbb{R} \times \mathbb{R}^2 \to \mathbb{R}^m$ is a random vector-valued function parametrized by the correlation parameter $\rho > 0$. We suppress the dependence of v^ρ on $\omega \in \Omega$.

Since v is vector-valued, our nonlinearity $\sigma : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ is matrix-valued. Let $\mathcal{H}_+^m$ denote the set of nonnegative-definite symmetric real $m \times m$ matrices, equipped with the metric induced by the Frobenius norm

$$|A|_F^2 := \mathrm{tr}(AA^T) = \mathrm{tr}(A^2)$$

Let σ belong to the space $\mathrm{Lip}(\mathbb{R}^m, \mathcal{H}_+^m)$.

Proposition 1.2.2. Show that $|\cdot|_F$ is a norm on $\mathcal{H}_+^m$.

Definiton 1.2.3. Given $\tau \geq 0$, we define the heat operator

$$\mathcal{G}_\tau v = G_\tau * v$$

where $G_\tau = (2\pi\tau)^{-1}\exp(-\frac{|x|^2}{2\tau})$ denotes the standard heat kernel. Define the spatially-smoothed noise $dW_t^\rho = G_\rho * dW_t$.

Proposition 1.2.3. We have

$$\mathbb{E}[dW_t^{\rho,i}(x)dW_{t'}^{\rho,i'}(x')] = \delta_{i,i'}\delta(t-t')G_{2\rho}(x-x')$$

Proof. □

Definiton 1.2.4. Define

$$L(\tau) = \ln(1+\tau) \quad \text{for } \tau \geq 0$$

and set

$$\gamma_\rho = \sqrt{\frac{4\pi}{L(1/\rho)}}$$

Definiton 1.2.5.

(Mild Solution 1)

A mild solution for (**??**) is a predictable random field v^ρ such that for all $s < t$, we have

$$v_t^\rho(x) = \mathcal{G}_{t-s}v_s^\rho(x) + \gamma_\rho\int_s^t \mathcal{G}_{t+\rho-r}[\sigma(v_r^\rho)dW_r](x)$$

which means

$$\begin{aligned} v_t^\rho(x) &= \mathcal{G}_{t-s}v_s^\rho(x) + \gamma_\rho\int_s^t\int G_{t+\rho-r}(y)\sigma(v_r^\rho)(x-y)dW_r(x-y)dy \\ &= \mathcal{G}_{t-s}v_s^\rho(x) + \gamma_\rho\int_s^t\int G_{t+\rho-r}(y)\sigma(v_r^\rho)(x-y)dW_r(x-y)dy \\ &= \mathcal{G}_{t-s}v_s^\rho(x) + \gamma_\rho\int\left(\int_s^t G_{t+\rho-r}(y)\sigma(v_r^\rho)(x-y)dW_r(x-y)\right)dy \end{aligned}$$

which can be interpreted as an Ito integral. We only look for the solution v_t^ρ in the spaces $\mathscr{X}_t^l$ of $\mathbb{R}^m$-valued random fields z on $\mathbb{R}^2$ that are $\mathcal{F}_t$-measurable and

$$||z||_l := \sup_{x\in\mathbb{R}^2}(\mathbb{E}|z(x)|^l)^{1/l} < \infty$$

Proposition 1.2.4. For any $l \geq 2$, there is a family of random operators $(\mathcal{V}_{s,t}^{\sigma,\rho})_{s<t}$ such that if $v_s \in \mathscr{X}_s^l$, then $v_t^\rho = \mathcal{V}_{s,t}^{\sigma,\rho}v_s$ is a mild solution of (**??**) for $t \geq s$. We often write $\mathcal{V}_t^{\sigma,\rho} := \mathcal{V}_{0,t}^{\sigma,\rho}$.

Shown by some standard fixed-point arguments.

Definiton 1.2.6. (Forward-backward SDE)

The system of SDE:

$$\begin{aligned} d\Gamma_{a,Q}^\sigma(q) &= J_\sigma(Q-q,\Gamma_{a,Q}^\sigma(q))dB(q), && a\in\mathbb{R}^m, 0<q<Q \\ \Gamma_{a,Q}^\sigma(0) &= a, && a\in\mathbb{R}^m, Q\geq 0 \\ J_\sigma(q,b) &= [\mathbb{E}\sigma^2(\Gamma_{a,Q}^\sigma(q))]^{1/2}, && q\geq 0, b\in\mathbb{R}^m \end{aligned}$$

for B a standard $\mathbb{R}^m$-valued Brownian motion and $A^{1/2}$ is the unique positive-definite matrix

square root of $A \in \mathcal{H}_+^m$.

Definiton 1.2.7. Given $\sigma \in \mathrm{Lip}(\mathbb{R}^m, \mathcal{H}_+^m)$, let $\overline{Q}_{\mathrm{FBSDE}}(\sigma) \in [0,\infty]$ be the supremum of all $Q \geq 0$ such that there is a continuous function $J_\sigma : [0,Q] \times \mathbb{R}^m \to \mathcal{H}_+^m$ satisfying the FBSDE and

$$\sup_{q \in [0,\overline{Q}]} \mathrm{Lip}(J_\sigma(q,\cdot)) < \infty$$

and define for $M, \beta \in (0,\infty)$

$$\Sigma(M,\beta) := \{\sigma \in \mathrm{Lip}(\mathbb{R}^m, \mathcal{H}_+^m) : |\sigma(u)|_F^2 \leq M + \beta^2 |u|^2 \text{ for all } u \in \mathbb{R}^m\}$$

Definiton 1.2.8.

(L^2-subcritical)

A nonlinearity σ is L^2-subcritical if $\sigma \in \Sigma(M,\beta)$ for some $M \in (0,\infty)$ and $\beta \in (0,1)$ and $\overline{Q}_{\mathrm{FBSDE}}(\sigma) > 1$.

1.3 Main Result

Theorem 1.3.1. Let σ be L^2-subcritical. Fix $Q \in (0,1)$ and define $\tilde{\sigma} := (1-Q)^{1/2} J_\sigma(Q,\cdot)$ and $\widetilde{\rho} := \rho^{1-Q}$. Then there is a new white noise $d\widetilde{W}$ such that $\mathcal{G}_{\widetilde{\rho}v}$ is an approximate mild solution of (**??**) with $(\widetilde{\rho}, \widetilde{\sigma}, d\widetilde{W})$ in place of (ρ, σ, dW).

Definiton 1.3.1. Denote $\mathcal{W}_2$ the Wasserstein-2 metric: for any two probability distributions μ, ν

$$\mathcal{W}_2(\mu,\nu) = \inf_\pi \left(\int |x-y|^2 \pi(dx,dy) \right)$$

where π has marginal distributions μ and ν.

Theorem 1.3.2. For each L^2-subcritical σ and $\overline{T} \in [1,\infty)$, there is a constant $C(\sigma,\overline{T}) \in (e,\infty)$ such that for all $v_0 \in L^\infty(\mathbb{R}^2,\mathbb{R}^m), t \in [\overline{T}^{-1},\overline{T}]$, and $\rho \in (0,C^1)$, the solution v^ρ of (**??**) satisfies

$$\mathcal{W}_2(v_t^{\rho(x)}, \Gamma_{a,1}^\sigma(1)) \leq C\langle ||v_0||_{L^\infty} \rangle \sqrt{\frac{\ln|\ln\rho|}{|\ln\rho|}}$$

where $(\Gamma_{a,1}^\sigma(q))_{q\in[0,1]}$ solves the FBSDE with $Q=1$ and $a = \mathcal{G}_t v_0(x)$.

2 Well-posedness of the FBSDE

2.1 Main Goal

The main result is

Theorem 2.1.1. Let $\sigma \in \mathrm{Lip}(\mathbb{R}^m, \mathcal{H}_+^m)$. For any $Q \in [0, \overline{Q}_{\mathrm{FBSDE}}(\sigma))$, we have

$$\overline{Q}_{\mathrm{FBSDE}}(\sigma) \geq Q + \mathrm{Lip}(J_\sigma(Q, \cdot))^{-2}$$

Definiton 2.1.1. $\mathcal{X}$ is the Banach space of $\mathcal{H}_+^m$-valued continuous functions on $\mathbb{R}^m$ with the norm (and this norm is finite)

$$||\sigma||_{\mathcal{X}} := \sup_{x \in \mathbb{R}^m} \frac{|\sigma(x)|_F}{\langle x \rangle}$$

Denote $\langle a \rangle := (|a|^2 + 1)^{1/2}$ the Japanese bracket.

Proposition 2.1.2. Prove $\mathcal{X}$ is a Banach space.

Proposition 2.1.3. $\mathrm{Lip}(\mathbb{R}^m; \mathcal{H}_+^m) \subset \mathcal{X}$.

2.2 SDE Solution Theory

Definiton 2.2.1. Given a $\mathbb{R}^m$-valued Brownian motion $(B(q))_{q \geq 0}$ adapted to a filtration $\{\mathscr{G}_q\}_{q \geq 0}$. For an adapted process Y on $[0, Q]$, a function $g : [0, Q] \times \mathbb{R}^m \to \mathcal{H}_+^m$ and a constant $a \in \mathbb{R}^m$, we define a new adapted process $\mathcal{R}_{a,Q}^g Y$ on $[0, Q]$ by

$$\mathcal{R}_{a,Q}^g Y(q) := a + \int_0^q g(Q - p, Y(p)) dB(q)$$

whenever this stochastic integral is defined. For $Q > 0$, define

$$\mathcal{A}_Q := \left\{ J : [0, Q] \times \mathbb{R}^m \to \mathcal{H}_+^m \text{ continuous} : \sup_{q \in [0,Q]} \mathrm{Lip}(J(q, \cdot)) < \infty \right\}$$

Proposition 2.2.1. Fix $L < \infty$ and $Q \in (0, \infty)$ and suppose that $g \in \mathcal{A}_Q$ satisfies

$$\sup_{q \in [0,Q]} \mathrm{Lip}(g(q, \cdot)) \leq L$$

Then, for any $a \in \mathbb{R}^m$, there is a unique strong solution $\Theta_{a,Q}^g$ to the SDE

$$\begin{aligned} d\Theta_{a,Q}^g(q) &= g(Q - q, \Theta_{a,Q}^g(q)) dB(q), \quad q \in [0, Q] \\ \Theta_{a,Q}^g(0) &= a \end{aligned}$$

The solution $\Theta_{a,Q}^g$ satisfies the moment bound

$$\sup_{q \in [0,Q]} \mathbb{E}|\Theta_{a,Q}^g(q)|^l < \infty \quad \text{for all } l \in [1, \infty)$$

Moreover, there exists a constant $C = C(L, Q)$ such that for any $Q' \in [0, Q]$ and any adapted

process Γ on $[0, Q']$, we have

$$\sup_{q\in[0,Q']} \mathbb{E}|\Gamma(q) - \Theta^g_{a,Q}(q)|^2 \leq C \sup_{[0,Q']} \mathbb{E}|\Gamma(q) - \mathcal{R}^g_{a,Q}\Gamma(q)|^2$$

and

$$\mathbb{E}|\Theta^g_{a,Q}(q) - \Theta^g_{\widetilde{a},Q}(q)|^2 \leq C|a - \widetilde{a}|^2$$

Proof. Define $\mathcal{K}_{L,Q'}$ on adapted processes on $[0, Q']$ by

$$\sup_{q\in[0,Q']} e^{-L^2 q} \left(\mathbb{E}|\Gamma(q)|^2\right)^{1/2},$$

which makes $\mathcal{R}^g_{a,Q}$ a contraction. □

2.3 Local Solution

Definiton 2.3.1. Let $Q > 0$ and let $\sigma \in \text{Lip}(\mathbb{R}^m; \mathcal{H}^m_+)$. We say that $J \in \mathcal{A}_Q$ is a *root decoupling function* for FBSDE on $[0, Q]$ if, for all $q \in [0, Q]$ and all $a \in \mathbb{R}^m$, we have

$$J(q, a) = [\mathbb{E}\sigma(\Theta^J_{a,q}(q))]^{1/2}$$

In this case, we also call J^2 the decoupling function. In the equation above, $\Theta^J_{a,q}$ is as in (**??**).

Definiton 2.3.2. For $\lambda \in (0, \infty)$, we define the set of functions

$$\Lambda(\lambda) := \{\sigma \in \text{Lip}(\mathbb{R}^m; \mathcal{H}^m_+) : \text{Lip}(\sigma) \leq \lambda\}$$

Proposition 2.3.1. Suppose that $\lambda \in (0, \infty), \sigma \in \Lambda(\lambda), Q_0 > 0$ and that J is a root decoupling function for FBSDE on $[0, Q_0]$. Then for all $Q \in [0, Q_0 \wedge \lambda^{-2})$, we have

$$\text{Lip}(J(Q, \cdot)) \leq (\lambda^{-2} - Q)^{-1/2}.$$

Proof. □

Lemma 2.3.2. Suppose that $c \in (0, \infty), \overline{Q} < c^{-2}$ and $f : [0, \overline{Q}] \to [c^2, \infty)$ satisfies

$$f(Q) \leq c^2 \exp\left\{\int_0^Q f(q)dq\right\}$$

for all $Q \in [0, \overline{Q}]$. Then

$$f(Q) \leq (c^{-2} - Q)^{-1} \quad \text{for all } Q \in [0, \overline{Q}]$$

Proof. Define $g(Q) = \int_0^Q f(q)dq$ and we have

$$g'(q) = f(q),$$

then

$$(1 - e^{-g(q)})' = g'(q)e^{-g(q)} = f(q)e^{-g(q)}$$

and hence

$$1 - e^{-g(Q)} = \int_0^Q f(q)e^{-g(q)}dq \leq c^2 Q$$

Therefore, we have

$$f(Q) \leq c^2 e^{g(Q)} \leq \frac{c^2}{1 - c^2 Q}$$

Here is another proof by (**??**), consider h defined on $[0, \overline{Q}]$ non-negative defined by

$$h(t) = \ln(f(t)/c^2),$$

then we have

$$h(t) \leq \int_0^t c^2 \exp(h(s))ds.$$

Define

$$G(x) = -e^{-x} + C$$

and we have

$$h(q) \leq -\ln(1 - c^2 q)$$

for any $q \in [0, \overline{Q}]$, which means

$$f(Q) \leq c^2/(1 - c^2 Q) = (c^{-2} - Q)^{-1}.$$

□

Definiton 2.3.3. Define

$$\mathcal{Q}_\sigma g(Q, a) = [\mathbb{E}\sigma^2(\Theta^g_{a,Q}(Q))]^{1/2},$$

where $(\Theta^g_{a,Q}(q))_{q\in[0,Q]}$ from (**??**). We note that a fixed point of Q_σ is a root decoupling function for (**??**). We also note that

$$|\mathcal{Q}_\sigma g(Q, a)|_F^2 = |\mathbb{E}\sigma^2(\Theta^g_{a,Q}(Q))]^{1/2}|_F^2 = \mathbb{E}\mathrm{tr}\sigma^2(\Theta^g_{a,Q}(Q)) = \mathbb{E}|\sigma(\Theta^g_{a,Q}(Q))|_F^2$$

Define the set of functions

$$\Lambda(M, \lambda) := \{\sigma \in \mathrm{Lip}(\mathbb{R}^m; \mathcal{H}^m_+) : \mathrm{Lip}(\sigma) \leq \lambda \text{ and } |\sigma(u)|_F^2 \leq M + \lambda^2|u|^2 \text{ for all } u \in \mathbb{R}^m\}$$

For $Q_0 < \lambda^{-2}$, define the set of functions $\mathcal{Z}_{Q_0,M,\lambda}$ by

$$\mathcal{Z}_{Q_0,M,\lambda} = \big\{g : [0, Q_0] \times \mathbb{R}^m \to \mathcal{H}^m_+ \text{ continuous},\\ g(q, \cdot) \in \Lambda((1 - \lambda^2 q)^{-2}M, (\lambda^{-2} - q)^{-1/2}) \text{ for all } q \in [0, Q_0]\big\}$$

We will construct the root decoupling function J_σ as a fixed point of Q_σ in a certain $\mathcal{Z}_{Q_0,M,\lambda}$.

Proposition 2.3.3. Fix $\lambda, M \in (0, \infty)$ and $Q_0 \in [0, \lambda^{-2})$. For any $\sigma \in \Lambda(M, \lambda)$, there is a unique root decoupling function $J_\sigma \in \mathcal{A}_{Q_0}$. In particular, we have

$$\overline{Q}_{\mathrm{FBSDE}}(\sigma) \geq \mathrm{Lip}(\sigma)^{-2}.$$

Moreover, there is a $C = C(Q_0, M, \lambda) < \infty$ such that for any $g \in \mathcal{Z}_{Q_0,M,\lambda}$, we have

$$\sup_{q\in[0,Q_0]} ||(g - J_\sigma)(q,\cdot)||_{\mathcal{X}} \le C \sup_{q\in[0,Q_0]} ||(g - \mathcal{Q}_\sigma g)(q,\cdot)||_{\mathcal{X}}$$

and indeed

$$\lim_{n\to\infty} \sup_{q\in[0,Q_0]} ||(\mathcal{Q}_\sigma^n g - J_\sigma)(q,\cdot)||_{\mathcal{X}} = 0$$

where Q_σ^n denotes the n-fold iterated application of Q_σ.

Proof. We know any decoupling function $J \in \mathcal{A}_{Q_0}$ has to be in $\mathcal{Z}_{Q_0,M,\lambda}$ for some $M', \lambda \in (0,\infty)$ □

2.4 Extension of the Solution

Lemma 2.4.1. Let $\sigma \in \mathrm{Lip}(\mathbb{R}^m; \mathcal{H}_+^m)$, whenever $0 \le Q_1 \le Q_2 < \overline{Q}_{\mathrm{FBSDE}}(\sigma)$, we have for any $b \in \mathbb{R}^m$,

$$J_\sigma^2(Q_2, b) = \mathbb{E}[J_\sigma^2(Q_1, \Gamma_{b,Q_2}^\sigma(Q_2 - Q_1))]$$

Proof. Remain. □

Lemma 2.4.2. Let $\sigma \in \mathrm{Lip}(\mathbb{R}^m; \mathcal{H}_+^m)$, whenever $0 \le Q_1 \le Q_2 < \overline{Q}_{\mathrm{FBSDE}}(\sigma)$ and if

$$Q_2 - Q_1 < \overline{Q}_{\mathrm{FBSDE}}(J_\sigma(Q_1,\cdot)),$$

we have

$$J_\sigma(Q_2, b) = J_{J_\sigma(Q_1,\cdot)}(Q_2 - Q_1, b) \quad \text{for all } b \in \mathbb{R}^m.$$

Proof. We have by (**??**) that,for any $Q \in [Q_1, \overline{Q}_{\mathrm{FBSDE}}(\sigma))$ and any $b \in \mathbb{R}^m$,

$$J_\sigma^2(Q, b) = \mathbb{E}[J_\sigma^2(Q_1, \Gamma_{b,Q}^\sigma(Q - Q_1))]$$

By assumption, there is a unique solution to the following FBSDE problem for $Q \in [0, Q_2 - Q_1]$:

$$\begin{aligned} d\Gamma_{b,Q}^{J_\sigma(Q_1,\cdot)}(q) &= J_{J_\sigma(Q_1,\cdot)}(Q - q, \Gamma_{b,Q}^{J_\sigma(Q_1,\cdot)}(q))dB(q), \quad q \in (0,Q) \\ \Gamma_{b,Q}^{J_\sigma(Q_1,\cdot)}(0) &= b \\ J_{J_\sigma(Q_1,\cdot)}(Q,b) &= (\mathbb{E}[J_\sigma^2(Q_1, \Gamma_{b,Q}^{J_\sigma(Q_1,\cdot)}(Q))])^{1/2} \end{aligned}$$

Also, if given $Q \in [0, \overline{Q}_{\mathrm{FBSDE}}(\sigma) - Q_1)$, (**??**) and (**??**) with Q replaced by $Q_1 + Q$ will yield

$$\begin{aligned} d\Gamma_{b,Q+Q_1}^\sigma(q) &= J_\sigma(Q + Q_1 - q, \Gamma_{b,Q+Q_1}^\sigma(q))dB(q), \quad q \in (0,Q) \\ \Gamma_{b,Q+Q_1}^\sigma(0) &= b \\ J_\sigma(Q + Q_1, b) &= (\mathbb{E}[\sigma^2(\Gamma_{b,Q+Q_1}^\sigma(Q))])^{1/2} = (\mathbb{E}[J_\sigma^2(Q_1, \Gamma_{b,Q+Q_1}^\sigma(Q))])^{1/2} \end{aligned}$$

which means ($\Gamma_{b,Q+Q_1}^\sigma(q)$ and $J_\sigma(Q + Q_1, b)$) will solve the previous FBSDE system, and by the uniqueness of FBSDE problem, we have

$$J_{J_\sigma(Q_1,\cdot)}(Q,b) = J_\sigma(Q + Q_1, b) \quad \text{for all } b \in \mathbb{R}^m \text{ and } Q \in [0, Q_2 - Q_1]$$

□

Proposition 2.4.3. For any $Q' \in [0, \overline{Q}_{\mathrm{FBSDE}}(\sigma))$, we have

$$\overline{Q}_{\mathrm{FBSDE}}(\sigma) \geq Q' + \overline{Q}_{\mathrm{FBSDE}}(J_\sigma(Q', \cdot))$$

Proof. For $Q' < \overline{Q}_{\mathrm{FBSDE}}\sigma$, there is a unique root decoupling function $J_\sigma \in \mathcal{A}_{Q'}$ for (**??**) on $[0, Q']$. Let $P \in [0, \overline{Q}_{\mathrm{FBSDE}}(J_\sigma(Q', \cdot)))$, there is a unique RDF $J_{J_\sigma(Q',\cdot)} \in \mathcal{A}_P$ in (**??**) with σ replaced by $J_\sigma(Q', \cdot)$.

We wish to extend the function J_σ to the time interval by outting □

3 Parabolic Equation for the decoupling function

3.1 Decoupling Flow

Definiton 3.1.1 (Parabolic Equation)**.**

The following parabolic equation

$$\begin{aligned}\partial_q H(q,b) &= \tfrac{1}{2}[H(q,b):\nabla_b^2]H(q,b)\\ H(0,b) &= \sigma^2(b)\end{aligned}$$

here for A, B matrices, we denote

$$A : B = \mathrm{tr}[AB]$$

and there is an explicit formula

$$([H(q,b):\nabla_b^2]H(q,b))_{ij} = \mathrm{tr}[H(q,b)\nabla_b^2]H_{ij}(q,b) = \sum_{k,l=1}^{m} H_{kl}(q,b)\frac{\partial^2 H_{ij}}{\partial b_k \partial b_l}(q,b)$$

Proof. We know

$$\mathrm{tr}[H(q,b)\nabla_b^2] = \sum_{k=1}^{n}\sum_{l=1}^{n} H_{kl}(q,b)\frac{\partial^2}{\partial b_k \partial b_l}$$

□

3.2 Main Theorem

Remark. The space

$$L^1(\mathcal{K}; W^{2,\infty}(U;\mathcal{H}_+^m))$$

means some functions which are L^1 with fixed space point and time varying and $W^{2,\infty}(U:\mathcal{H}_+^m)$ with fixed time point and space varying, with the norm

$$||f|| = \int ||f(t,\cdot)||_{W^{2,\infty}} dt.$$

For two normed spaces, their intersection will be equipped with a norm obtained by adding both the norms together.

Definiton 3.2.1. An *almost classical solution* to (**??**) on a time interval $[0, Q_0)$ is a continuous function $H : [0, Q_0) \times \mathbb{R}^m \to \mathcal{H}_+^m$ such that the following conditions hold:

1. For every compact $\mathcal{K} \subset (0, Q_0)$ and bounded open $U \subset \mathbb{R}^m$,

   $$H_{\mathcal{K}\times U} \in L^1(\mathcal{K}; W^{2,\infty}(U;\mathcal{H}_+^m)) \cap C^1(\mathcal{K}\times U;\mathcal{H}_+^m).$$

   where $W^{2,\infty}(U;\mathcal{H}_+^m)$ is the Sobolev space of functions on U taking values in $\mathcal{H}_+^m$ with weak second derivative mearuable ans essentially bounded.

2. We have $H(0,b) = \sigma^2(b)$ for all $b \in \mathbb{R}^m$.

3. We have

$$\partial_q H(q,b) = \tfrac{1}{2}[H(q,b) : \nabla_b^2] H(q,b) \quad \text{for almost all } (q,b) \in (0,Q_0) \times \mathbb{R}^m$$

Proposition 3.2.1. Fix $Q_0 \in (0, \overline{Q}_{\mathrm{FBSDE}}(\sigma))$. Suppose H is an almost classical solution of (**??**) on $[0,Q_0]$ such that $\sup_{q\in[0,Q_0]} \operatorname{Lip}\sqrt{H(q,\cdot)}$. Also assume that for each compact $\mathcal{K} \subset (0,Q_0)$, there exists a constant $C(\mathcal{K}) < \infty$ such that for all $R > 0$,

$$||H|_{\mathcal{K}\times U_R}||_{L^1(\mathcal{K};W^{2,\infty}(U;\mathcal{H}_+^m))\cap C^1(\mathcal{K}\times U;\mathcal{H}_+^m)} \le CR^C.$$

Then $\sqrt{H} = J_\sigma$ on $[0,Q_0] \times \mathbb{R}^m$.

Theorem 3.2.2. Suppose $\sigma \in \operatorname{Lip}(\mathbb{R};\mathbb{R}^+)$ and $I := \{b \in \mathbb{R} : \sigma(b) > 0\}$ is an open interval. Suppose also that there exists $\beta \in (0,\infty)$ such that the applicable following condition holds.

(i) If I is a bounded interval, then there is a constant $K \in [1,\infty)$ such that for all $x \in I$, we have

$$K^{-1} \operatorname{dist}(x, I^c) \le \sigma(x) \le \beta \operatorname{dist}(x, I^c).$$

(ii) If $I = \mathbb{R}$, then there are constants $K \in [1,\infty)$ and $\gamma \in [0,2]$ such that for all $x \in \mathbb{R}$, we have

$$K^{-1}\langle x\rangle^{\gamma/2} \le \sigma(x) \le (\beta^2|x|^2 + K)^{1/2} \wedge (K\langle x\rangle^{\gamma/2}).$$

(iii) If I is the half-line (a,∞) or (∞,a), then there are constants $K \in [1,\infty)$ and $\gamma[0,2]$ such that for all $x \in I$, we have

$$K^{-1}(|x-a| \wedge |x-a|^{\gamma/2}) \le \sigma(x) \le (\beta|x-a| \wedge (K|x-a|^{\gamma/2})).$$

Then we have $\overline{Q}_{\mathrm{FBSDE}}(\sigma) \ge \beta^{-2}$.

Corollary 3.2.3. Let $\sigma \in \operatorname{Lip}(\mathbb{R};\mathbb{R}_{\ge 0})$ and suppose there exist $\beta \in (0,\infty)$ and $k \in (0,\beta]$ such that $k|b| \le \sigma(b) \le \beta|b|$ for all $b \in \mathbb{R}$. Then $\overline{Q}_{\mathrm{FBSDE}}(\sigma) \ge \beta^{-2}$.

Proof. By the previous theorem, since $\sigma(b) \ge k|b| > 0$ for any nonzero b, so we may know that for $(0,\infty)$, let $\gamma = 2, K = k^{-1} \vee 1 \vee \beta$ and then we know

$$K^{-1}|b| \le k|b| \le \sigma|b| \le \beta|b| \le \beta|b| \wedge K|b|$$

and it is similar for $(-\infty,0)$ and we are done. □

4 SPDE

5 Open Problem

One possible question is that if σ is bounded, then in term of the solution u_ϵ of the stochastic pde

$$\partial_t u_\epsilon = \tfrac{1}{2}\Delta u_\epsilon + \sigma(u)dW^\epsilon,$$

what will u_ϵ converges to as $\epsilon \to 0$.

5.1 Limit behavior of the parabolic equation

In this part, we consider a quasi-linear parabolic equation

$$u_t = \tfrac{1}{2}uu_{xx}, \quad t \in [0,\infty), x \in \mathbb{R}$$

and according to the paper by Alex.D and Cole.G, we have such conclusions about the existence and regularity of the solution of this equation. Firstly, let us define the weak solution for the equation

Definiton 5.1.1 (Weak solution)**.** Given $T > 0$, we say that $u \in \mathcal{C}\big([0,T); W^{1,\infty}_{\text{loc}}(\mathbb{R})\big)$ is a *weak solution* of the parabolic equation on $[0,T)$ if $u \geq 0$ and for all $\psi \in C_c^\infty([0,T)\times\mathbb{R})$,

$$\int_{\mathbb{R}} u_0\psi(0,\cdot) + \int_{[0,T)\times\mathbb{R}} \left[u\,\partial_t\psi - \frac{1}{2}\,\partial_x(\psi u)\,\partial_x u\right] = 0. \tag{1}$$

With better regularity of u and $\partial_x^2 u$, we have

Definiton 5.1.2 (Strong solution)**.** Given $\overline{T} > 0$, a weak solution of the parabolic equation on $[0,\overline{T})$ is called *strong* if for every $T \in (0,\overline{T})$ we have

$$u \in \langle x\rangle^2 L^\infty([0,T]\times\mathbb{R}) \qquad \text{and} \qquad \partial_x^2 u \in L^1\big([0,T]; L^\infty(\mathbb{R})\big).$$

where

$$\langle x\rangle = \sqrt{(x^2+1)}$$

is the Japanese braacket.

For merely weak solution, we have a lemma to secure the smoothness

Lemma 5.1.1. Let u be a weak solution of the parabolic equation on a time interval $[0,T)$. Then on the set $\{(t,x) \in (0,T)\times\mathbb{R} | u(t,x) > 0\}$, u is qualitatively smooth and satisfies the parabolic equation pointwise.

By the way, the equation admits some kind of comparsion principle

Proposition 5.1.2. Given $T > 0$, suppose $u^\pm \in \mathcal{C}_{\text{loc}}([0,T]\times\mathbb{R})$ satisfy $\partial_x^2 u^\pm \in L^1_t L^\infty_x$ and $\partial_t u \in \langle x\rangle^2 L^1_t L^\infty_x$ as well as

$$\partial_t u^+ \geq \tfrac{1}{2}u^+\partial_x^2 u^+, \quad \partial_t u^- \leq \tfrac{1}{2}u^-\partial_x^2 u^-, \quad a.e.$$

Then if $u^-(0,\cdot) \leq u^+(0,\cdot)$, we have $u^- \leq u^+$.

5.1.1 Uniform convergence

Now we would like to consider for some initial condition with relatively good properties, if the limit of u at the time of infinity will exist and what it is, and a basic observation is that $u(t,\cdot)$ will approach to some constant function on $\mathbb{R}$ in general, so the we have a claim

Theorem 5.1.3 (Limit Behavior for Periodic IC)**.** Assume $u \in \mathcal{C}_{\text{loc}}([0,\infty)\times\mathbb{R})$ is a weak solution of the parabolic equation, with a positive bounded initial condition $u_0 = u(0,\cdot) > 0$ and $\partial_x^2 u \in L^1_{\text{loc};t}L^\infty_x$, and u_0 is periodic. Then

$$\lim_{t\to\infty} u(t,x) = C(u_0) \quad \text{for a.e. } x \in \mathbb{R},$$

where $C(u_0)$ is some constant only related to the initial condition.

Theorem 5.1.4 (Limit Behavior for Step-like IC)**.** Assume $u \in \mathcal{C}_{\text{loc}}([0,\infty)\times\mathbb{R})$ is a weak solution of the parabolic equation, with the initial condition u_0 which satisfies that there exists a function $f \in \overline{C^\infty}^u$ with a compact support and a constant $c > 0$, such that

$$u_0 = f \text{ on some bounded interval } [a,b], \quad u_0 = \begin{cases} f(a) & \text{on } [-\infty, a] \\ f(b) & \text{on } [b, +\infty] \end{cases}$$

and $\partial_x^2 u \in L^1_{\text{loc};t}L^\infty_x$. Then

$$\lim_{t\to\infty} u(t,x) = C(u_0) \quad \text{for a.e. } x \in \mathbb{R},$$

where $C(u_0)$ is some constant only related to the initial condition.

Lemma 5.1.5. For a weak solution u of the parabolic equation, with $\partial_x^2 u \in L^1_{\text{loc};t}L^\infty_x$, if there exists two constant m, M such that $m \le u_0 : u(0,\cdot) \le M$, then $m \le u \le M$ on $[0,\infty)\times\mathbb{R}$.

Proof. Let M be u^+ and since u is a weak solution, so it is continuous automatically by the lemma. And apply the comparision principle pointwisely, we know $u \le M$ for every point contained in a compact subset of $[0,\infty)\times\mathbb{R}$ and hence every point. The same argument may be applied to m. □

Now let us prove the case of periodic IC.

Proof. By the lemma above, we may know that there exists $m, M > 0$ such that $m \le u \le M$ and hence the equation is not degenerated at every point.

Notice here $\partial_t u, \partial_x u, \partial_x^2 u$ are all defined naturally and even smooth by the smoothness of weak solution, so we may consider the energy estimation for u. Define

$$E(t) = \int_{-L}^{L} (\partial_x u(t,y))^2 dy$$

and then by compactness of $[-L, L]$ and DCT, we have

$$\begin{aligned}\frac{dE(t)}{dt} &= \int_{-L}^{L} 2\partial_{xt}u(t,y)\partial_x u(t,y)dy \\ &= 2\left(u_t(t,x)u_x(t,x)|_{-L}^{L} - \int_{-L}^{L} u_t(t,y)\partial_{xx}u(t,y)dy\right) \\ &= -\int_{-L}^{L} u(t,y)(\partial_{xx}u(t,y))^2 dy \leq 0\end{aligned}$$

in both cases. So $u(t,\cdot)$ is bounded in $H^1([-L,L])$. Notice $E(t) \geq 0$ and

$$\frac{dE(t)}{dt} \leq -m\int_{-L}^{L}(\partial_{xx}u(t,y))^2 dy \leq 0$$

which means we may choose a subsequence u_{t_n} such that $\|\partial_x^2 u(t_n,\cdot)\|_{L^2([-L,L])} \leq 1/n$. By Morrey's inequality, $u(t_n,\cdot), \partial_x(t_n,\cdot)$ are both compactly embedded in $\mathcal{C}^{0,1/2}([-L,L])$ and hence we may choose a further subsequence $u(t_{n_j},\cdot) \to u_\infty$ with $\partial_x(t_{n_j},\cdot) \to u'_\infty$ in $\mathcal{C}^{0,1/2}([-L,L])$ for some u_∞ and u'_∞ and $u'_\infty = \partial_x u_\infty$ in weak sense. Therefore, $u_\infty \in W^{1,\infty}([-L,L])$.

We consider $\partial_x^2 u_\infty$ only as a distribution, then for any $\phi \in C_c^\infty([-L,L])$, we have

$$\begin{aligned}\langle \partial_t u(t_n,\cdot), \phi\rangle &= \tfrac{1}{2}\int_{[-L,L]} u(t_n,\cdot)\partial_x^2 u(t_n,\cdot)\phi \\ &= \tfrac{1}{2}\left(\int_{[-L,L]} \partial_x(u(t_n,\cdot)\partial_x u(t_n,\cdot)) - (\partial_x u(t_n,\cdot))^2\phi\right) \\ &= \tfrac{1}{4}\int_{[-L,L]} \partial_x^2(u^2(t_n,\cdot))\phi - \tfrac{1}{2}\int_{[-L,L]}(\partial_x u(t_n,\cdot))^2\phi \\ &= \tfrac{1}{4}\int_{[-L,L]} u^2(t_n,\cdot)\partial_x^2\phi - \tfrac{1}{2}\int_{[-L,L]}(\partial_x u(t_n,\cdot))^2\phi\end{aligned}$$

since $u_\infty, \partial_x u_\infty$ are both continuous function on $[-L,L]$ and $u(t_n,\cdot) \to u_\infty, \partial_x u(t_n,\cdot) \to \partial_x u_\infty$ uniformly, we have $u^2(t_n,\cdot) \to u_\infty^2, (\partial_x u(t_n,\cdot))^2 \to (\partial_x u_\infty)^2$ uniformly. Therefore we have

$$\begin{aligned}\langle \partial_t u(t_n,\cdot), \phi\rangle &= \tfrac{1}{4}\int_{[-L,L]} u^2(t_n,\cdot)\partial_x^2\phi - \tfrac{1}{2}\int_{[-L,L]}(\partial_x u(t_n,\cdot))^2\phi \\ &\to \tfrac{1}{4}\int_{[-L,L]} u_\infty^2\partial_x^2\phi - \tfrac{1}{2}\int_{[-L,L]}(\partial_x u_\infty)^2\phi \\ &= \tfrac{1}{2}\langle u_\infty\partial_x^2 u_\infty, \phi\rangle\end{aligned}$$

where $u_\infty\partial_x^2 u_\infty$ is well-defined since u_∞ is positive continuous function, which may be approached by smooth function. Notice

$$\|\partial_t u(t_n,\cdot)\|_{L^2(-L,L)} \leq \tfrac{1}{m}\|\partial_x^2 u(t_n,\cdot)\|_{L^2(-L,L)} \to 0$$

and hence $vv_\infty = 0$ as a distribution, which means

$$\partial_x^2(u_\infty^2) = 2(\partial_x u_\infty)^2 \in \mathcal{C}([-L,L])$$

(Check) We consequently have $\partial_x^2 u_\infty \in L^\infty([-L,L])$ by some direct computation. Now $u_\infty \partial_x^2 u_\infty$ may be interpreted by $L^\infty([-L,L])$ functions and then $\partial_x^2 u_\infty = 0$ by u_∞ positive, so u_∞ is some constant between m and M when u_0 is periodic, and the constant is exactly c as we defined in another case. We denote this constant as u_∞ uniformly for the following.

Now we may consider the truncation solution U_T which is the solution of the parabolic equation with the initial condition $u(T,\cdot)$, by the comparision principle, we know U_T will be bounded in $[u_\infty - \epsilon, u_\infty + \epsilon]$ if $|u(T,\cdot) - u_\infty|_u < \epsilon$. To sum up, let T to be t_n we chose before and then we will know $u(t+t_n, x) = U_T(t,x) \in [u_\infty - \epsilon_n, u_\infty + \epsilon_n]$ for any $t \geq 0$ and $\epsilon_n \to 0$ with $n \to \infty$, so we obtained that

$$\lim_{t\to\infty} u(t,\cdot) = u_\infty$$

in $\mathcal{C}(\mathbb{R})$, and in particular, for any $x \in \mathbb{R}$,

$$\lim_{t\to\infty} u(t,x) = u_\infty.$$

□

Lemma 5.1.6. For a weak solution u, with positve lower bound and upper bound, of the parabolic equation and smooth initial condition u_0, with $\partial_x^2 u \in L^1_{\text{loc};t} L^\infty_x$, if there are limits

$$\lim_{x\to-\infty} u_0(x), \quad \lim_{x\to+\infty} u_0(x)$$

exist, then $\lim_{x\to\infty} \partial_x u(t,x) = 0$.

Proof. We want to show that

$$\lim_{x\to-\infty} u(x,t) = \lim_{x\to-\infty} u_0(x), \quad \lim_{x\to+\infty} u(x,t) = \lim_{x\to\infty} u_0(x).$$

By applying Gaussian estimation for $v = u - \lim_{x\to-\infty} u_0(x)$, which satisfies a uniformly elliptic equation

$$v_t = \frac{u}{2}\partial_x^2 v$$

and we have for any $t > 0$,

$$|v(t,x)| \leq \int \frac{C}{t^{n/2}} \exp\left(-\frac{C'|x-y|^2}{t}\right) v_0(y)dy \to 0, \quad \text{if } x \to -\infty$$

for some positive constants C, C' and $v_0 = v(0,\cdot)$. Therefore, we know for any $t > 0$, we have

$$\lim_{x\to-\infty} u(x,t) = \lim_{x\to-\infty} v(x,t) + \lim_{x\to-\infty} u_0(x) = \lim_{x\to-\infty} u_0(x)$$

and the other equality can be shown in an analogous argument.

Now since the initial condition is smooth with postive lower and upper bound, we may use the Schauder estimation for u and hence for any $0 < \alpha < 1, T > 0$, we have on $[0,T] \times \mathbb{R}$

$$\|u\|_{\mathcal{C}_b^{2+\alpha,1+\alpha/2}} \leq C(\|u_0\|_{\mathcal{C}_b^{2+\alpha}}, m, M, T)$$

and hence $\sup_{t\in[0,T]} \|\partial_x^2 u\|_{L^\infty} < \infty$. Then for any $t \leq T$ we know $\lim_{x\to\infty} \partial_x u = 0$ since

$\partial_x u(t,\cdot)$ is Lipschitz. $\square$

The case for bump IC is a little different, here is the proof.

Proof. We use the same notation of m, M as the proof in periodic case. Assume $u_0 \in C^\infty$ at first.

For the energy estimation, we consider

$$E_L(t) = \int_{-L}^{L} (\partial_x u(t,y))^2 dy$$

and $E(t) := E_\infty(t)$, then we may know that

$$\begin{aligned}\frac{dE_L(t)}{dt} &= 2u_t(t,x)\partial_x u(t,x)|_{-L}^{L} - \int_{-L}^{L} u(t,y)(\partial_x^2 u(t,y))^2 dy \\ &= u(t,x)\partial_x(t,x)\partial_x^2 u(t,x)|_{-L}^{L} - \int_{-L}^{L} u(t,y)(\partial_x^2 u(t,y))^2 dy\end{aligned}$$

and since $\|\partial_x^2 u(t,\cdot)\|_{L^\infty}$(should be guanranteed by some other estimation) is bound on $[0,T]$ for any T. By the lemma above, we know

$$\lim_{L\to\infty} \frac{dE_L(t)}{dt} = -\int u(t,y)(\partial_x^2 u(t,y))^2 dy$$

for almost every t. On the other hand, notice

$$\frac{dE_L(t)}{dt} \le u(t,x)\partial_x(t,x)\partial_x^2 u(t,x)|_{-L}^{L}$$

so for $T > 0$, we may choose $L_{n;T}$ such that $u(t,x)\partial_x(t,x)\partial_x^2 u(t,x)|_{-L}^{L} \le 1/n$ if $L > L_{n;T}$ and hence $\dfrac{dE_{L_{n;T}}(t)}{dt} \le 1/n$ for any $t \in [0,T]$. Then

$$E_{L_{n:T}(t)} \le E_{L_{n;T}(0)} + T/n \le E(0) + T/n$$

for any $t \in [0,T]$, and let $n \to \infty$, we may know $E(t) \le E(0)$ for any $t > 0$. Moreover, $E(t)$ is decreasing. For any $\delta > 0$, if then are T_δ such that for any $t > T_\delta$, $\|\partial_x^2 u(t,\cdot)\|_{L^2} > \delta$, then we know for $L > L_{n;MT_\delta}$

$$\frac{dE_L(t)}{dt} = 1/n - I_L(t)$$

where $I_L(t) := \int_{-L}^{L} u(t,y)(\partial_x^2 u(t,y))^2 dy$ and hence

$$E_L(MT_\delta) \le E_L(T_\delta) - \left(\int_{T_\delta}^{MT_\delta} I_L(t)dt - (M-1)T_\delta/n\right)$$

so we may choose M sufficiently large such that $\frac{(M-1)}{2} T_\delta m\delta > 2E(0)$, and choose n large enough such that $1/n < m\delta/4$ and L large enough such that

$$\int_{-L}^{L}\int_{T_\delta}^{MT_\delta} u(t,y)(\partial_x^2 u(t,y))^2 dydt = \int_{T_\delta}^{MT_\delta} I_L(t)dt > 3m(M-1)T_\delta\delta/4$$

since $\int\int_{T_\delta}^{MT_\delta} u(t,y)(\partial_x^2 u(t,y))^2 dydt > m(M-1)T_\delta\delta$ by Fubini and hence L can be chosen by MCT. Then we know that $E_{L_{n;MT_\delta}(MT_\delta)} \leq E(0) - 2E(0) < 0$, which will be a contradiction, so we know we may choose a subsequence $u(t_n, \cdot)$ such that

$$\|\partial_x^2 u(t_n, \cdot)\|_{L^2} \to 0$$

Therefore, we may use the analog argument for periodic case to show that u_∞ will be a constant.

Now let show that the conclusion is true for any $f \in \overline{C^\infty}^u$ with compact support. For any $\epsilon > 0$, we may consider that ϕ_ϵ to be some smooth function with compact support and $\|\phi_\epsilon - f\| < \epsilon$, then we know

$$\phi_\epsilon - \epsilon + c < u_0 = f + c < \phi_\epsilon + \epsilon + c.$$

Consider the weak solutions $\phi_\epsilon^-, \phi_\epsilon^+$ given by $\phi_\epsilon - \epsilon + c, \phi_\epsilon + \epsilon + c$. By the comparsion principle,

$$\phi_{\epsilon,\infty}^- < \phi_{\epsilon,\infty}^+, \quad \phi_{\epsilon,\infty}^- > \phi_{\epsilon,\infty}^+ - 2\epsilon, \quad \phi_{\epsilon,\infty}^- \leq \lim_{t\to\infty} \inf_x u \leq \lim_{t\to\infty} \sup_x u \leq \phi_{\epsilon,\infty}^+$$

and it is easy to check that

$$u_\infty = \limsup_{\epsilon\to 0} \phi_{\epsilon,\infty}^-$$

□

5.1.2 The value of Weak Solution at Infinity Time

Before discussing the decay of the spatial derivative, we record the value of the limiting constant in the two model cases.

Theorem 5.1.7 (Limit Value for Periodic IC)**.** In the periodic initial condition case, assume that u_0 has period L and satisfies $0 < m \leq u_0 \leq M$. Then the limiting constant is

$$u_\infty = \exp\left(\frac{1}{L}\int_0^L \ln u_0(y) dy\right).$$

Proof. We consider the value

$$\Phi(t) = \int_0^L \ln(u(t,y)) dy.$$

By the equation and periodicity,

$$\frac{d\Phi}{dt} = \int_0^L u^{-1}(t,y) u_t(t,y) dy = \frac{1}{2}\int_0^L u_{yy}(t,y) dy = \frac{1}{2} u_y(t,y)\Big|_0^L = 0.$$

Hence $\Phi(t) = \Phi(0)$. Since we already know that $u(t,\cdot)$ converges uniformly to a constant u_∞, the dominated convergence theorem gives

$$\int_0^L \ln u_0(y) dy = \lim_{t\to\infty} \Phi(t) = L \ln u_\infty.$$

This proves the formula. □

Theorem 5.1.8 (Limit Value for Bump IC)**.** In the bump case, assume that there exists a constant $c > 0$ and a compact set $K \subset \mathbb{R}$ such that

$$u_0(x) = c, \qquad x \in \mathbb{R} \setminus K.$$

Then the limiting constant is $u_\infty = c$.

Proof. This is the same limit-at-infinity argument proved above for the step-like initial condition with the same left and right limit. More precisely, one first compares the solution with periodic extensions on $[-L, L]$; their limiting constants are

$$\exp\left(\frac{1}{2L}\int_{-L}^{L} \ln u_0^{(L)}(y)dy\right),$$

and these constants converge to c as $L \to \infty$. If the bump is not one-sided, we split it into the upper and lower perturbations

$$h = (u_0 - c)^+ + c, \qquad g = c - (u_0 - c)^-.$$

Then $g \le u_0 \le h$, and the two comparison solutions with initial data g and h both have limiting constant c. The comparison principle therefore gives

$$c \le \liminf_{t\to\infty} \inf_x u(t,x) \le \limsup_{t\to\infty} \sup_x u(t,x) \le c.$$

Hence $u_\infty = c$. □

5.1.3 Decaying rate of the spatial derivative for period case

Now we turn to the main point. We would like to show that, for the two classes of initial condition above, the spatial derivative has at most the order t^{-1} at large time. In this part we assume the initial condition has enough regularity so that the solution is classical for $t > 0$, and we keep the uniform positive bounds

$$0 < m \le u(t,x) \le M < \infty. \tag{2}$$

The constants below may depend on m, M, the regularity norm of u_0, and the period or the size of the compact perturbation, but not on t.

Theorem 5.1.9 (Derivative Decay for Periodic IC)**.** Assume that u_0 is positive, bounded, sufficiently smooth, and L-periodic. Then there is a constant $C > 0$ such that, for all $t \ge 1$,

$$\|u_x(t,\cdot)\|_{L^\infty(\mathbb{T}_L)} \le \frac{C}{t}.$$

In fact, u_x decays exponentially in time.

Proof.

Let $p(t,x) := \partial_x u(t,x)$, and then since u is smooth, we have $\partial_t p = \frac{1}{2}\partial_x(u\partial_x p)$. Since u is periodic, p has zero spatial average on $\mathbb{T}_L$, where L is a period length for u:

$$\int_{\mathbb{T}_L} p(t,x)dx = 0.$$

Define

$$E(t) = \int_{\mathbb{T}_L} p(t,x)^2 dx.$$

and by integration by parts, we will have

$$\begin{aligned}\frac{d}{dt}E(t) &= 2\int_{\mathbb{T}_L} p\partial_t p dx = \int_{\mathbb{T}_L} p\partial_x(u\partial_x p)dx \\ &= -\int_{\mathbb{T}_L} u(\partial_x p)^2 dx \\ &\le -m\int_{\mathbb{T}_L} (\partial_x p)^2 dx.\end{aligned}$$

By the Poincare inequality (**??**) on $\mathbb{T}_L$, because p has zero average,

$$\int_{\mathbb{T}_L} (\partial_x p)^2 dx \ge \left(\frac{2\pi}{L}\right)^2 \int_{\mathbb{T}_L} p^2 dx.$$

and hence

$$E'(t) \le -m\left(\frac{2\pi}{L}\right)^2 E(t).$$

By Gronwall's inequality,

$$\|\partial_x u(t,\cdot)\|_{L^2(\mathbb{T}_L)} \le \exp\left(-\frac{1}{2}m\left(\frac{2\pi}{L}\right)^2 t\right)\|\partial_x u(0,\cdot)\|_{L^2(\mathbb{T}_L)}.$$

Notice that u is bounded and apply (**??**), for $t \ge 1$,

$$\|p(t,\cdot)\|_{L^\infty(\mathbb{T}_L)} \le C\|p(t-1,\cdot)\|_{L^2(\mathbb{T}_L)}.$$

for uniform constant C_0. Therefore

$$\|\partial_x u(t,\cdot)\|_{L^\infty(\mathbb{T}_L)} \le C_0 e^{-ct}$$

for some $c > 0$. Since $e^{-ct} \le C_1 t^{-1}$ for $t \ge 1$ for some $C_1 > 0$, we obtain

$$\|\partial_x u(t,\cdot)\|_{L^\infty(\mathbb{T}_L)} \le \frac{C}{t}.$$

where $C = C_0 C_1$. □

5.1.4 Decaying rate of the spatial derivative for bump case

For the bump initial condition, the Poincare inequality is no longer available. A basic idea is to first obtain the coarse order $t^{-1/2}$, and then use the convergence $u \to c$ and a parabolic scaling argument to improve it to t^{-1}. We will prove the $t^{-1/2}$ decaying rate for the equation

at first.

Theorem 5.1.10 (Derivative decaying rate of t-root for Bump IC)**.** Assume that u_0 is positive, bounded, smooth and satisfies

$$u_0 - c \in C_c^{2+\alpha}(\mathbb{R})$$

for some $c > 0$ and $0 < \alpha < 1$. Let u be the classical solution of (**??**). Then there is a constant $C > 0$ such that, for all sufficiently large t,

$$\|\partial_x u(t,\cdot)\|_{L^\infty(\mathbb{R})} \leq \frac{C}{\sqrt{t}}.$$

Proof. Let $p := u_x$ and then we know the equation that p will satify is

$$p_t = \frac{1}{2}(up_x)_x$$

Consider a scaled solution of the equation above, that is for $\tau > 0$, for $f : \mathbb{R}^+ \times \mathbb{R} \to \mathbb{R}$, define $f^\tau(x,t) = f(\tau x, \tau^2 t)$ and since

$$(p^\tau)_t = \tau^2 (p_t)^\tau, \quad (u^\tau (p^\tau)_x)_x = (u^\tau)_x (p^\tau)_x + u^\tau (p^\tau)_{xx} = \tau^2 [(up_x)_x]^\tau$$

we have p^τ is a solution of

$$v_t = \frac{1}{2}(u^\tau v_x)_x$$

Then by Theorem.1 from [**?**], the equation above admits a fundamental solution g_τ on $[0,1] \times \mathbb{R}$ with bound

$$0 \leq g_\tau(t,x;\tau,\xi) \leq K\gamma(t-\tau, x-\xi)$$

where K, γ only depend on the bound of u^τ which is the same for all the $\tau > 0$, so we know

$$\begin{aligned} |p^\tau(1,x)| &\leq K \int_{\mathbb{R}} \gamma(1, x-y)|p^\tau(0,y)|dy \\ &\leq K\|\gamma(1,\cdot)\|_{L^\infty(\mathbb{R})}\|p^\tau(0,\cdot)\|_{L^1(\mathbb{R})} \end{aligned}$$

and notice that

$$\|p^\tau(0,\cdot)\|_{L^1(\mathbb{R})} = \tfrac{1}{\tau}\|p(0,\cdot)\|.$$

Therefore, we will have

$$|p(\tau^2,\cdot)|_{L^\infty(\mathbb{R})} \leq C\frac{1}{\sqrt{\tau^2}}\|p(0,\cdot)\|_{L^1(\mathbb{R})}$$

for any $\tau > 0$ and the desired rate of decaying is obtained. □

Theorem 5.1.11 (L^1-control of the prabolic equation)**.** With the same assumption with (**??**), we will have

$$\sup_{t\geq 0}\|u(t,\cdot) - c\|_{L^1(\mathbb{R})} \leq C\|u(0,\cdot) - c\|_{L^1(\mathbb{R})}$$

Proof. Let us show that for any $t > 0$, $(u(t,\cdot) - c) \in L^1(\mathbb{R})$ at first. According to [**?**], for any T, the equation will admit a fundamental solution g admit a Gaussian upper bound γ

with coefficients only related to T, so for $t < T$, we know that

$$\begin{aligned}|u(t,x) - c| &\le K \int_{\mathbb{R}} \gamma(x-y)|w(y)|dy \\ &= K \int_{\text{supp}(w)} \gamma(x-y)|w(y)|dy \\ &\le K' \|w\|_{L^1(\mathbb{R})} \exp(-\beta x^2)\end{aligned}$$

for some constant $K, K', \beta > 0$ where $w := u(0,\cdot) - c$, which means $u(t,\cdot) \in L^1(\mathbb{R})$ for any $t < T$ and hence $u(t,\cdot)$ is L^1 integrable for any $t \ge 0$.

We still will use the above bound with the coefficient related to t and the following calculation is done for fixed t. For $\epsilon > 0$, for $\phi_\epsilon(x) = \sqrt{x^2+\epsilon^2} - \epsilon$ and we will have

$$|\phi_\epsilon(x) - |x|| = \frac{2|x|\epsilon}{\sqrt{x^2+\epsilon^2} + \epsilon + |x|} \le \frac{|x|\epsilon}{\sqrt{x^2+\epsilon^2}}.$$

Consider the logarithmic quantity of

$$q(t,x) = \ln \frac{u(t,x)}{c}$$

and we know q satisfy the equation

$$q_t = \tfrac{1}{2} u_{xx}$$

Also, since u is bounded, we know $|q|$ is equivalent with $|u-c|$, and we may apply the similar Gaussian bound to $|q|$. Then we know

$$\begin{aligned}&\int_{\mathbb{R}} |\phi_\epsilon(q(t,x)) - q(t,x)|dx \\ &\le \int_{\mathbb{R}} \frac{|q(t,x)|\epsilon}{\sqrt{q(t,x)^2+\epsilon^2}} dx \\ &\le \int_{|q(t,x)|>\epsilon} \epsilon dx + \int_{|q(t,x)|<\epsilon} K'' \|w\|_{L^1(\mathbb{R})} \exp(-\beta x^2) dx \\ &\le \epsilon \frac{2}{\sqrt{\beta}} \sqrt{-\ln\left(\frac{\epsilon}{K''\|w\|_{L^1(\mathbb{R})}}\right)} + I_\epsilon \to 0\end{aligned}$$

for some constant $K'' > 0$, with $\epsilon \to 0$ where $I_\epsilon := \int_{|u(t,x)|<\epsilon} K''\|w\|_{L^1(\mathbb{R})} \exp(-\beta x^2)dx \to 0$ by DCT, so we have $\phi_\epsilon(q(t,x)) \to |q(t,x)|$ in $L^1(\mathbb{R})$ for any $t \ge 0$.

Now, consider

$$\begin{aligned}\frac{d}{dt}\int_{\mathbb{R}} \phi_\epsilon(q(t,x))dx &= \frac{1}{2}\int_{\mathbb{R}} \phi_\epsilon'(q(t,x)) u_{xx}(t,x)dx \\ &= -\frac{1}{2}\int_{\mathbb{R}} \phi_\epsilon''(q(t,x))(uq_x^2)(t,x)dx \le 0\end{aligned}$$

and hence

$$\int_{\mathbb{R}} \phi_\epsilon(q(t,\cdot)) \le \int_{\mathbb{R}} \phi_\epsilon(q(0,\cdot))$$

Therefore, we pass the limit and obtain

$$\int_{\mathbb{R}} |q(t,\cdot)| \le \int_{\mathbb{R}} |q(0,\cdot)|$$

for any $t > 0$ and hence

$$\|u(t,\cdot) - c\|_{L^1(\mathbb{R})} \le C\|w\|_{L^1(\mathbb{R})}$$

for some constant C and we are done. □

Theorem 5.1.12 (Derivative Decay for Bump IC)**.** With the same assumption with (**??**), the decaying rate is $1/t$, i.e. there is a constant $C > 0$ such that, for all sufficiently large t,

$$\|\partial_x u(t,\cdot)\|_{L^\infty(\mathbb{R})} \le \frac{C}{t}.$$

Proof. Let

$$w(t,x) := u(t,x) - c, \qquad A(t,x) := \frac{1}{2}u(t,x).$$

Then w satisfies the linear non-divergence form equation

$$w_t = A(t,x)w_{xx}, \tag{3}$$

and by (**??**),

$$\frac{m}{2} \le A(t,x) \le \frac{M}{2}.$$

By (**??**), we have

$$\|u_x(t,\cdot)\|_{L^\infty(\mathbb{R})} \le Kt^{-1/2}. \tag{4}$$

In particular,

$$\|A_x(t,\cdot)\|_{L^\infty(\mathbb{R})} \le Kt^{-1/2}. \tag{5}$$

for some constant $K > 0$. Now we use the equation (**??**). The coefficient A is uniformly elliptic. Moreover, by the coarse estimate (**??**), after the parabolic scaling

$$A_R(\tau,z) := A(R\tau, x_0 + \sqrt{R}z), \qquad 1 \le \tau \le 2,$$

we have

$$\|\partial_z A_R\|_{L^\infty} = \sqrt{R}\|A_x\|_{L^\infty([R,2R]\times\mathbb{R})} \le C.$$

Thus the spatial Dini modulus of the rescaled coefficient is uniformly controlled. Applying the Gaussian upper bound for non-divergence form parabolic equations with spatial Dini control gives, for the fundamental solution Γ of (**??**),

$$0 \le \Gamma(t,x;s,y) \le C(t-s)^{-1/2}\exp\left(-\frac{|x-y|^2}{C(t-s)}\right)$$

whenever $R \le s < t \le 2R$, with C independent of R. Taking $s = t/2$ and using the

representation formula,

$$\begin{aligned}|w(t,x)| &\le \int_{\mathbb{R}} \Gamma(t,x;t/2,y)|w(t/2,y)|dy \\ &\le Ct^{-1/2}\|w(t/2,\cdot)\|_{L^1(\mathbb{R})}.\end{aligned}$$

By (**??**),

$$\|u(t,\cdot)-c\|_{L^\infty(\mathbb{R})} = \|w(t,\cdot)\|_{L^\infty(\mathbb{R})} \le Ct^{-1/2}. \tag{6}$$

Step 3: scaling and Schauder improvement. Fix a large time t_0 and a point $x_0 \in \mathbb{R}$. Let

$$r := \frac{\sqrt{t_0}}{4}$$

and define

$$W(\tau,z) := w(t_0 + r^2\tau, x_0 + rz), \qquad (\tau,z) \in Q_1 := (-1,0]\times(-1,1).$$

Then W satisfies

$$W_\tau = A^{(t_0,x_0)}(\tau,z)W_{zz},$$

where

$$A^{(t_0,x_0)}(\tau,z) := A(t_0 + r^2\tau, x_0 + rz).$$

Since $t_0 + r^2\tau \in [15t_0/16, t_0]$, the ellipticity constants are uniform. In the spatial direction,

$$\partial_z A^{(t_0,x_0)} = rA_x(t_0 + r^2\tau, x_0 + rz),$$

and by (**??**),

$$\|\partial_z A^{(t_0,x_0)}\|_{L^\infty(Q_1)} \le C.$$

Also, by standard interior regularity for the original equation,

$$\|A_t(t,\cdot)\|_{L^\infty(\mathbb{R})} \le Ct^{-1}$$

for large t. Hence

$$\|\partial_\tau A^{(t_0,x_0)}\|_{L^\infty(Q_1)} = r^2\|A_t\|_{L^\infty} \le C.$$

Therefore the coefficient $A^{(t_0,x_0)}$ has a uniform $C^{\alpha/2,\alpha}$ bound on Q_1. Applying the interior Schauder estimate to $W_\tau = A^{(t_0,x_0)}W_{zz}$ gives

$$|W_z(0,0)| \le C\|W\|_{L^\infty(Q_1)}.$$

Since

$$W_z(0,0) = rw_x(t_0,x_0) = ru_x(t_0,x_0),$$

we obtain

$$|u_x(t_0,x_0)| \le Cr^{-1}\|W\|_{L^\infty(Q_1)}.$$

By (**??**),

$$\|W\|_{L^\infty(Q_1)} \le Ct_0^{-1/2}.$$

Since $r^{-1} = 4t_0^{-1/2}$, it follows that

$$|u_x(t_0, x_0)| \leq Ct_0^{-1}.$$

Taking the supremum over $x_0 \in \mathbb{R}$ proves

$$\|u_x(t_0, \cdot)\|_{L^\infty(\mathbb{R})} \leq Ct_0^{-1}.$$

This completes the proof. □

6 Convergence of the Heat Flow with Bounded Nonlinearility

In this section, we are going to show that for the smoothened nonlinear stochastic heat equation on $\mathbb{R}^+ \times \mathbb{R}$

$$dv_t^\rho = \tfrac{1}{2}\Delta v_t^\rho + \mathcal{G}_\rho\left[\sigma(v_t^\rho)dW_t\right] \tag{SHEBD}$$

as an analog of

$$dv_t^\rho = \tfrac{1}{2}\Delta v_t^\rho + \gamma_\rho\mathcal{G}_\rho\left[\sigma(v_t^\rho)dW_t\right] \tag{SHE}$$

will admit a Gaussian limit, that is

$$v_t^\rho \Rightarrow \mathcal{N}(0, \sigma_\infty)$$

for some constant σ_∞ related to some Lipschitz bounded nonlinear function σ. To analyze the convergence of the (**??**), we define

$$\sigma_\rho := \gamma_\rho^{-1}\sigma$$

and the equation (**??**) can be written equivalently as

$$dv_t^\rho = \tfrac{1}{2}\Delta v_t^\rho + \gamma_\rho\mathcal{G}_\rho\left[\sigma_\rho(v_t^\rho)dW_t\right] \tag{SHETB}$$

In this section, without specifying, σ_ρ is the rescaling of a fixed σ as above.

6.1 Main Conjecture

Let us recall the main theorem of [**?**]

Theorem 6.1.1. For each L^2-subcritical σ and $\overline{T} \in [1, \infty)$, there is a constant $C(\sigma, \overline{T})$ such that for all $v_0 \in L^\infty(\mathbb{R}^2; \mathbb{R}^m)$, $t \in [\overline{T}^{-1}, \overline{T}], x \in \mathbb{R}^2$ and $\rho \in (0, C^{-1})$, the solution v^ρ of (**??**) satisfies ‘

$$\mathcal{W}_2(v_t^\rho(x), \Gamma_{a,1}^\sigma(1)) \le C\langle\|v_0\|_{L^\infty}\rangle\sqrt{\frac{\ln|\ln\rho|}{|\ln\rho|}}$$

where $(\Gamma_{a,1}^\sigma(q))_{q\in[0,1]}$ solves the FBSDE

$$\begin{aligned} d\Gamma_{a,Q}^\sigma(q) &= J_\sigma(Q-q, \Gamma_{a,Q}^\sigma(q))dB(q), && a \in \mathbb{R}^m, 0 < q < Q \\ \Gamma_{a,Q}^\sigma(0) &= a, && a \in \mathbb{R}^m, Q \ge 0 \\ J_\sigma(q,b) &= [\mathbb{E}\sigma^2(\Gamma_{a,Q}^\sigma(q))]^{1/2}, && q \ge 0, b \in \mathbb{R}^m \end{aligned} \tag{FBSDE}$$

with $Q = 1$ and $a = \mathcal{G}_t\left[v_0\right](x)$.

According to the theorem, we may know that the distribution of v_t^ρ and $\Gamma_{a,1}^\sigma(1)$ are almost the same for ρ small enough and t large enough, so we expect the same conclusion for (**??**), and if $C(\sigma_\rho, \overline{T})$ can be bounded for fixed $\overline{T}$, then the distribution of the limit solution of (**??**) may be approximated by weak limit of $\Gamma_{a,1}^{\sigma_\rho}$ when $\rho \to 0$.

Recall some scaling conclusion of J_σ

Proposition 6.1.2. For $\sigma \in \mathrm{Lip}(\mathbb{R}^m; \mathcal{H}^m_+)$ and $\zeta > 0$, we have

$$\overline{Q}_{\mathrm{FBSDE}}(\zeta\sigma) = \zeta^{-2}\overline{Q}_{\mathrm{FBSDE}}(\sigma)$$

and for all $q \in [0, \zeta^{-2}\overline{Q}_{\mathrm{FBSDE}}(\sigma)]$ and $b \in \mathbb{R}^m$,

$$J_\zeta(q, b) = \zeta J_\sigma(\zeta^2 q, b)$$

By the theorem, we may know

$$\begin{aligned}\Gamma^{\sigma_\rho}_{a,1}(q) &= \int_0^q J_{\sigma_\rho}(1-r, \Gamma^{\sigma_\rho}_{a,1}(r))dB(r) \\ &= \gamma_\rho^{-1}\int_0^q J_\sigma(\gamma_\rho^{-2}(1-r), \Gamma^{\sigma_\rho}_{a,1}(r))dB(r)\end{aligned}$$

and for each $r < 1$, if $v_0 \in L^\infty(\mathbb{R}^2; \mathbb{R})$, then we know $J_\sigma(t, \cdot)$ will converge to some constant uniformly according to some parabolic equation analysis, so with $\rho \to \infty$, we expect the distribution $\gamma_\rho v^\rho_t(x)$ will coincide with $\gamma_\rho \Gamma^{\sigma_\rho}_{a,1}(q)$ whose weak limit is Browinian motion with mean σ_∞.

6.2 First Stage Approximation

Definiton 6.2.1. For a bounded positive Lipschitz function σ, define σ_ρ as $\gamma_\rho^{-1}\sigma$.

The scaled time parameter is defined by $L_\rho(q) = L(q)/L(1/\rho)$.

The space $\mathscr{H}^l_t$ is consisted of $\mathbb{R}$-valued random fields z on $\mathbb{R}^2$ that are $\mathcal{F}_t$-measurable with

$$\|z\|_l := \sup_{x\in\mathbb{R}^2}(\mathbb{E}|z(x)|^l)^{1/l} < \infty$$

and $\langle a\rangle := \sqrt{(|a|^2+1)}$ is the Japanese bracket.

Give $\zeta > 0$ and $k \in \mathbb{R}^2$, the square

$$B_{\zeta,k} := \zeta \cdot (k + [0,1)^2)$$

and for $x, z \in \mathbb{R}^2$, let $k_{\zeta,z}(x)$ be the unique integral vector $k \in \mathbb{Z}^2$ such that $x \in B_{\zeta,k+z}(x)$. Given a field f on $\mathbb{R}^2$, $\zeta \in (0,\infty)$ and $z \in \mathbb{R}^2$, define

$$\mathcal{S}_{\zeta,z}f(x) := \frac{1}{\zeta^2}\int_{B_{\zeta,k_{\zeta,z}(x)+z}} f(y)dy; \quad \mathcal{S}_\zeta := \mathcal{S}_{\zeta,0}.$$

For $a \in \mathbb{R}$ and $q \geq 0$ and a Lipschitz function σ, define

$$J_{\sigma,\rho}(q,a) = \left[\mathbb{E}\sigma^2\left(\mathcal{V}^{\sigma,\rho}_{T_\rho(q)}a(x)\right)\right]^{1/2}$$

where $S_\rho(q) := L_\rho(q/\rho)$ and $T_\rho = S_\rho^{-1}$. The operator $\mathcal{V}^{\sigma,\rho}$ is defined as that if $v_s \in \mathscr{H}^l_s$ for $l \geq 2$, then $v^\rho_t := \mathcal{V}^{\sigma,\rho}_{s,t}v_s$ is a mild solution of

$$du^\rho_t = \frac{1}{2}\Delta u^\rho_t + \gamma_\rho\sigma(u^\rho_t)dW^\rho_t$$

which means $\mathcal{V}_{s,t}^{\sigma_\rho,\rho} v_s$ is a mild solution of

$$du_t^\rho = \frac{1}{2}\Delta u_t^\rho + \sigma(u_t^\rho) dW_t^\rho$$

with $s \geq t$. We often write $\mathcal{V}_{0,t}^{\sigma,\rho} := \mathcal{V}_t^{\sigma,\rho}$.

For a standard Brownian motion $(B_q)_{q\geq 0}$ adapted to $(\mathcal{F}_q)_{q\geq 0}$. For an adapted process Y on $[0,Q]$, a function $g : [0,Q] \times \mathbb{R} \to \mathcal{H}_+$ and a constant a, define

$$\mathcal{R}_{a,Q}^g Y(q) := a + \int_0^q g(Q-p, Y(p)) dB_p$$

and

$$\mathcal{A}_Q := \left\{ J : [0,Q] \times \mathbb{R} \to \mathcal{H}_+ \text{ continuous: } \sup_{q\in[0,Q]} \mathrm{Lip}(J(q,\cdot)) < \infty \right\}$$

for $Q > 0$.

For some $g \in \mathcal{A}_Q$, and $a \in \mathbb{R}$, there is a unique strong solution $\Theta_{a,Q}^g$ to the SDE

$$\begin{aligned} d\Theta_{a,Q}^g(q) &= g(Q-q, \Theta_{a,Q}^g)(q) dB_q, \quad q \in [0,Q] \\ \Theta_{a,Q}^g(0) &= a \end{aligned}$$

with a moment bound

$$\sup_{q\in[0,Q]} \mathbb{E}|\Theta_{a,Q}^g(q)|^l < \infty \text{ for all } l \geq 1$$

Moreover, for any $Q' \in [0,Q]$ and any adapated process Γ on $[0,Q']$, we have

$$\sup_{q\in[0,Q']} \mathbb{E}|\Gamma(q) - \Theta_{a,Q}^g(q)|^2 \leq C(\sup \mathrm{Lip}(g), Q) \sup_{q\in[0,Q']} \mathbb{E}|\Gamma(q) - \mathbb{R}_{a,Q}^g \Gamma(q)|^2$$

and

$$\mathbb{E}|\Theta_{a,Q}^g(q) - \Theta_{a',Q}^g(q)| \leq C|a-a'|^2$$

with the same constant C.

$J \in \mathcal{A}_Q$ is a root decoupling function on $[0,Q]$ if for all $q \in [0,Q], a \in \mathbb{R}$ we have

$$J(q,a) = [\mathbb{E}\sigma^2(\Theta_{a,Q}^J(q))]^{1/2}$$

which is a fixed point of

$$Q_\sigma : \mathcal{A}_Q \to \mathcal{A}_Q, \quad Q_\sigma g(Q,a) = \left[\mathbb{E}\sigma^2(\Theta_{a,Q}^g(Q))\right]^{1/2}$$

Theorem 6.2.1. Let σ to be a Lipschitz function with $\mathrm{Lip}(\sigma) \leq \lambda$, and $(v_t)_{t\geq 0}$ solve

$$v_t(x) = \mathcal{G}_{t-s}[v_s](x) + \int_s^t \mathcal{G}_{t+\rho-r}[\sigma(v_r^\rho) dW_r](x)$$

with initial condition $v_0 \in \mathscr{H}_0^2$. Let $T = \alpha\rho, \iota \in [0,1]$ and $\xi \in [0, \sqrt{T/L(1/\rho)}]$, $x, z \in \mathbb{R}^2$,

where $\alpha < \frac{8}{3}\left(e^{4\pi/\lambda^2}-1\right)$. Then we have for $t \in [\xi^2 L(1/\rho)^{-\beta}, T], \iota_1, \iota_2 \in [0,1], \beta \in [0,1)$

$$\mathbb{E}\left|\left(\mathcal{S}_{\xi,z}[\sigma_\rho^2 \circ v_t](x)\right)^{1/2} - J_{\sigma_\rho,\rho}(S_\rho(L(1/\rho)^{\iota_1}\xi^2), \mathcal{G}_{L(1/\rho)^{\iota_2}\xi^2}\left[v_t\right](x))\right|^2 \le CL(1/\rho)^{1+\beta}$$

for some constant C when $\rho < C^{-1}$.

Proof. In this proof, we denote $\mathcal{V}^{\sigma_\rho,\rho}$ as $\mathcal{V}$. Let $R = B_{\xi,k_{\xi,z}(x)+z}$ and $\tau_0 = t/4, \tau_2 := t$ and $\tau_1 = (\tau_0+\tau_2)/2$.

Define

$$\begin{aligned}
E_1 :&= \mathbb{E}\left|\left(\frac{1}{\xi^2}\int_R \sigma_\rho^2(v_t(y))dy\right)^{1/2} - \left(\frac{1}{\xi^2}\int_R \sigma_\rho^2(\mathcal{V}_{\tau_1,\tau_2}\mathcal{Z}_X\mathcal{G}_{\tau_1-\tau_0}\left[v_{\tau_0}(y)\right])dy\right)^{1/2}\right|^2\\
E_2 :&= \mathbb{E}\left|\left(\frac{1}{\xi^2}\int_R \sigma_\rho^2(\mathcal{V}_{\tau_1,\tau_2}\mathcal{Z}_X\mathcal{G}_{\tau_1-\tau_0}\left[v_{\tau_0}(y)\right])dy\right)^{1/2} - J_{\sigma_\rho,\rho}(S_\rho(\tau_2-\tau_1), \mathcal{G}_{\tau_1-\tau_0}\left[v_{\tau_0}\right](X))\right|^2\\
E_3 :&= \mathbb{E}\left|J_{\sigma_\rho,\rho}(S_\rho(\tau_2-\tau_1), \mathcal{G}_{\tau_1-\tau_0}\left[v_{\tau_0}\right](X)) - J_{\sigma_\rho,\rho}(S_\rho(L(1/\rho)^{\iota_1}\xi^2), \mathcal{G}_{L(1/\rho)^{\iota_2}\xi^2}\left[v_t\right](x))\right|^2
\end{aligned}$$

For E_1, we have

$$\begin{aligned}
E_1 &\le \frac{1}{\xi^2}\int_R \mathbb{E}\left|\sigma_\rho(\mathcal{V}_{\tau_0,\tau_2}v_{\tau_0}(y)) - \sigma_\rho(\mathcal{V}_{\tau_1,\tau_2}\mathcal{Z}_X\mathcal{G}_{\tau_1-\tau_0}\left[v_{\tau_0}(y)\right])\right|^2 dy\\
&\le \gamma_\rho^{-2}\lambda^2\left(\frac{1}{\xi^2}\int_R \mathbb{E}\left|\mathcal{V}_{\tau_0,\tau_2}v_{\tau_0}(y) - \mathcal{V}_{\tau_1,\tau_2}\mathcal{Z}_X\mathcal{G}_{\tau_1-\tau_0}\left[v_{\tau_0}(y)\right]\right|^2 dy\right)\\
&\le \gamma_\rho^{-2}\lambda^2\|v_{\tau_0}\|_2^2\left(\frac{|\sigma|_\infty L_\rho(\tau_1)}{4\pi(1-\lambda^2 L_\rho(\tau_2-\tau_1))} + \frac{\xi^2}{4\tau_1(1-\lambda^2 L_\rho(\tau_2-\tau_1)/4\pi)}\right.\\
&\left.+\frac{\tau_2-\tau_1}{\tau_1}\left(\frac{1}{1-\lambda^2 L_\rho(\tau_2-\tau_1)/4\pi} + \frac{\lambda^2}{8\pi(1-\lambda^2 L_\rho(\tau_2-\tau_1)/4\pi)^2}\right)\right)\\
&\le C_1' L(1/\rho)(C_1' + C_1' L(1/\rho)^\beta + C_1') \le C_1 L(1/\rho)^{1+\beta}
\end{aligned}$$

for some constant C_1, C_1'.

For E_2, we consider $R_i, i \in \mathcal{I}$ to be a squares partition pf R with edge length ξ/N, and $\hat{R}_i$ is the square with the same center with R_i but whose edge length $3\xi/N$. Denote $\mathcal{G}_{\tau_1-\tau_0}\left[v_{\tau_0}\right](X) =: a(x)$ and $i(x)$ to be the index in $\mathcal{I}$ such that $x \in R_{i(x)}$,

$$\begin{aligned}
&\mathbb{E}\left|\left(\frac{1}{\xi^2}\int_R \sigma_\rho^2(\mathcal{V}^{\hat{R}_{i(x)}}_{\tau_1,\tau_2}a(x))dx\right)^{1/2} - \left(\frac{1}{\xi^2}\int_R \sigma_\rho^2(\mathcal{V}_{\tau_1,\tau_2}a(x))dx\right)^{1/2}\right|^2\\
&\le \gamma_\rho^{-2}\lambda^2\left(\frac{1}{\xi^2}\int_R \mathbb{E}\left|\mathcal{V}^{\hat{R}_{i(x)}}_{\tau_1,\tau_2}a(x) - \mathcal{V}_{\tau_1,\tau_2}a(x)\right|^2 dx\right)\\
&\le \gamma_\rho^{-2}\frac{C'\lambda^2 S_\rho(T)}{4\pi - \lambda^2 L(T/\rho)}\Xi_{2;\tau_1,\tau_2}[a]^2\\
&\le C_2'\Xi_{2;\tau_1,\tau_2}[a]^2
\end{aligned}$$

for some fixed constant C' and some constant C_2' only related to λ. Let

$$X_i := \frac{1}{(\xi/N)^2}\int_{R_i}\sigma_\rho^2(\mathcal{V}^{\hat{R}_i}_{\tau_1,\tau_2}a(x))dx$$

and we will have

$$EX_i = J^2_{\sigma_\rho,\rho}(S_\rho(\tau_2-\tau_1),a).$$

by (5.45) in [**?**], we have

$$\begin{aligned}
&\mathbb{E}\left|\left(\frac{1}{\xi^2}\int_R \sigma_\rho^2(\mathcal{V}^{\hat{R}_{i(x)}}_{\tau_1,\tau_2}a(x))dx\right)^{1/2} - J_{\sigma_\rho,\rho}(S_\rho(\tau_2-\tau_1),a)\right|^2 \\
=&----\mathbb{E}\left|\left(N^{-2}\sum_{i\in\mathcal{I}}X_i\right)^{1/2} - (\mathbb{E}X_i)^{1/2}\right|^2 \le C''\Xi_{2;\tau_1,\tau_2}[a]^2
\end{aligned}$$

for some fixed constant C''. Notice by Propoisition 4.4 in [**?**], we know

$$\begin{aligned}
\Xi_{2;\tau_1,\tau_2}[a]^2 &\le \|a\|_2^2 + S_\rho(\tau_2-\tau_1)\gamma_\rho^{-2}|\sigma|^2_\infty \\
&= \|\mathcal{G}_{\tau_1-\tau_0}\left[v_{\tau_0}\right](X)\|_2^2 + L_\rho(\tau_2-\tau_1)|\sigma|^2_\infty/4\pi \\
&\le \|v_{\tau_0}\|_2^2 + L_\rho(\tau_2-\tau_1)|\sigma|^2_\infty/4
\end{aligned}$$

and notice that

$$\|v_{\tau_0}\|_2^2 \le \|v_0\|_2^2 + L_\rho(\tau_0)|\sigma|^2_\infty/4\pi$$

Therefore, we have

$$E_2 \le C_2'(\|v_0\|_2^2 + 2L(\alpha)|\sigma|^2_\infty/4) \le C_2$$

for some constant C_2 related to σ, v_0 and α.

For E_3, we have

$$\begin{aligned}
&\frac{1}{2}\mathbb{E}\left|J_{\sigma_\rho,\rho}(S_\rho(\tau_2-\tau_1),\mathcal{G}_{\tau_1-\tau_0}\left[v_{\tau_0}\right](X)) - J_{\sigma_\rho,\rho}(S_\rho(L(1/\rho)^{\iota_1}\xi^2),\mathcal{G}_{L(1/\rho)^{\iota_2}\xi^2}\left[v_t\right](x))\right|^2 \\
&\quad\le \mathbb{E}\left|J_{\sigma_\rho,\rho}(S_\rho(L(1/\rho)^{\iota_1}\xi^2),\mathcal{G}_{\tau_1-\tau_0}\left[v_{\tau_0}\right](X)) - J_{\sigma_\rho,\rho}(S_\rho(L(1/\rho)^{\iota_1}\xi^2),\mathcal{G}_{L(1/\rho)^{\iota_2}\xi^2}\left[v_t\right](x))\right|^2 \\
&\quad+ \mathbb{E}\left|J_{\sigma_\rho,\rho}(S_\rho(\tau_2-\tau_1),\mathcal{G}_{\tau_1-\tau_0}\left[v_{\tau_0}\right](X)) - J_{\sigma_\rho,\rho}(S_\rho(L(1/\rho)^{\iota_1}\xi^2),\mathcal{G}_{\tau_1-\tau_0}\left[v_{\tau_0}\right](X))\right|^2 \\
&\quad\le \mathbb{E}|\mathcal{G}_{\tau_1-\tau_0}\left[v_{\tau_0}\right](X) - \mathcal{G}_{L(1/\rho)^{\iota_2}\xi^2}\left[v_t\right](x)|^2/(4\pi\lambda^{-2}/L(1/\rho) - L_\rho(L(1/\rho)^{\iota_1}\xi^2)/L(1/\rho)) \\
&\quad+ \sup_{x\in\mathbb{R}^2}\mathbb{E}|\sigma_\rho(\mathcal{V}_p\mathcal{G}_{\tau_1-\tau_0}\left[v_{\tau_0}\right](X)) - \sigma_\rho(\mathcal{V}_{p-q,p}\mathcal{G}_{\tau_1-\tau_0}\left[v_{\tau_0}\right](X))|^2 := E_1' + E_2'
\end{aligned}$$

where $p = ((\tau_2-\tau_1)\vee L(1/\rho)^{\iota_1}\xi^2), q = ((\tau_2-\tau_1)\wedge L(1/\rho)^{\iota_1}\xi^2)$. We may estimate E_1' by

$$\begin{aligned}
E_1' &\le 2L(1/\rho)C_3'(\|v_{\tau_0}\|_2^2 + \|v_t\|_2^2) \\
&\le 2L(1/\rho)C_3'(2\|v_0\|_2^2 + L_\rho(\tau_0)|\sigma|^2_\infty/4\pi + L_\rho(t)|\sigma|^2_\infty/4\pi) \\
&\le 4L(1/\rho)C_3'(2\|v_0\|_2^2 + L(\alpha)|\sigma|^2_\infty/4\pi)
\end{aligned}$$

and for E_2',

$$\begin{aligned}
E_2' &\le \frac{L(1/\rho)\lambda^2}{4\pi}\|\mathcal{V}_p\mathcal{G}_{L(1/\rho)^{\iota_1}\xi^2}\left[v_t\right](X) - \mathcal{V}_{p-q,p}\mathcal{G}_{L(1/\rho)^{\iota_1}\xi^2}\left[v_t\right](X)\|_2^2 \\
&\le \frac{L(1/\rho)\lambda^2}{4\pi}\frac{ML_\rho(p-q)}{4\pi(1-\lambda^2L_\rho(p))} \\
&\le C_3''L(1/\rho)
\end{aligned}$$

for some constant C_3'' and to sum up we have

$$E_3 \leq C_3 L(1/\rho)$$

for some constant C_3.a □

6.3 Turning off the noise and flattening

In this section, we assume $|\sigma| < M$.

Proposition 6.3.1. Suppose $\lambda \in (0, \infty)$, $\mathrm{Lip}(\sigma) \leq \lambda, \rho > 0$, and $\tau_0 \leq \tau_1 < \tau_2$ satisfy $\tau_2 - \tau_1 < a$. For all $v_{\tau_0} \in \mathscr{H}^2_{\tau_0}$, we have

$$\|\mathcal{V}^\rho_{\tau_0,\tau_2} v_{\tau_0} - \mathcal{V}^\rho_{\tau_1,\tau_2} \mathcal{G}_{\tau_1-\tau_0}[v_{\tau_0}]\|_2^2 \leq \frac{ML(\tau_1/\rho)}{4\pi(1-\lambda^2 L(a/\rho))}$$

Proof. Assume without loss of generality that $\tau_0 = 0$. For $t \geq \tau_1$, define $v_t = \mathcal{V}_{0,t} v_0$ and $\widetilde{v}_t = \mathcal{V}_{\tau_1,t} \mathcal{G}_{\tau_1}[v_0]$. We have

$$v^\rho_t - \widetilde{v}^\rho_t = \int_0^{\tau_1} \mathcal{G}_{t+\rho-r}[\sigma(v_r) dW_r] + \int_{\tau_1}^t \mathcal{G}_{t+\rho-r}[(\sigma(v_r) - \sigma(\widetilde{v}_r)) dW_r]$$

Then by Ito isometry we will have

$$\begin{aligned}
\mathbb{E}|(v^\rho_t - \widetilde{v}^\rho_t)(x)|^2 =& \int_0^{\tau_1} \int G^2_{t+\rho-r}(x-y) \mathbb{E}|\sigma(v_r(y))|^2 dy dr \\
&+ \int_{\tau_1}^t \int G^2_{t+\rho-r}(x-y) \mathbb{E}|\sigma(v_r(y)) - \sigma(\widetilde{v}_r(y))|^2 dy dr \\
\leq & M \int_0^{\tau_1} \int G^2_{t+\rho-r}(x-y) dy dr + \lambda^2 \int_{\tau_1}^t \frac{\|v^\rho_r - \widetilde{v}^\rho_r\|_2^2}{t+\rho-r} dr \\
\leq & \frac{M}{4\pi} L(\tau_1/\rho) + \lambda^2 \int_{\tau_1}^t \frac{\|v^\rho_r - \widetilde{v}^\rho_r\|_2^2}{t+\rho-r} dr
\end{aligned}$$

Let $f(\tau) = \|v^\rho_{\tau_1+\tau} - \widetilde{v}^\rho_{\tau_1+\tau}\|_2^2$ for $\tau \in [0, \tau_2 - \tau_1]$ and we will have

$$f(\tau) \leq A + B \int_{\tau_1}^\tau \frac{f(r)}{\tau+\rho-r} dr$$

which will gives for $F := \sup_{\tau \in [0, \tau_2-\tau_1]} f(\tau)$,

$$\begin{aligned}
f(\tau) &\leq A + B \int_{\tau_1}^\tau \frac{F}{\tau+\rho-r} dr \\
\implies f(\tau) &\leq A + BL\left(\frac{\tau-\tau_1}{\rho}\right) F
\end{aligned}$$

Apply supremum related to τ on the both side and we obtain

$$F \leq A + BL\left(\frac{\tau_2-\tau_1}{\rho}\right) F \implies F \leq \frac{A}{1 - BL((\tau_2-\tau_1)/\rho)}$$

and we are done. □

Proposition 6.3.2. Suppose $\lambda \in (0,\infty)$, $\mathrm{Lip}(\sigma) \le \lambda, \rho > 0$, and $\tau_0 \le \tau_1 < \tau_2$ satisfy $\tau_2 - \tau_1 < a$. For all $x, X \in \mathbb{R}^2$ and all $v_{\tau_0} \in \mathscr{H}^2_{\tau_0}$, we have

$$\mathbb{E}|(\mathcal{V}^\rho_{\tau_1,\tau_2}\mathcal{G}_{\tau_1-\tau_0}[v_{\tau_0}] - \mathcal{V}^\rho_{\tau_1,\tau_2}\mathcal{Z}_X\mathcal{G}_{\tau_1-\tau_0}[v_{\tau_0}])(x)|^2 \le \frac{2\pi(2a+1)\|v_0\|_2^2}{\tau_1(4\pi - \lambda^2(L(a/\rho)+a))}(1+|X-x|)^2$$

Proof. Assume $\tau_0 = 0$ and for all $\tau \in [0, \tau_2 - \tau_1]$, we have

$$\begin{aligned}
&(\mathcal{V}^\rho_{\tau_1,\tau_1+\tau}\mathcal{G}_{\tau_1}[v_0] - \mathcal{V}^\rho_{\tau_1,\tau_1+\tau}\mathcal{Z}_X\mathcal{G}_{\tau_1}[v_0])(x)\\
=&\int [G_{\tau_1+\tau}(x-y) - G_{\tau_1}(X-y)]v_0(y)dy\\
+&\int_0^\tau \mathcal{G}_{\tau+\rho-r}\left[(\sigma(\mathcal{V}_{\tau_1,\tau_1+r}\mathcal{G}_{\tau_1}[v_0]) - \sigma(\mathcal{V}_{\tau_\infty,\tau_\infty+\nabla}\mathcal{Z}_X\mathcal{G}_{\tau_1}[v_0]))dW_{\tau_1+r}\right](x)
\end{aligned}$$

Take second moments, we have

$$\begin{aligned}
&\mathbb{E}\left|(\mathcal{V}^\rho_{\tau_1,\tau_1+\tau}\mathcal{G}_{\tau_1}[v_0] - \mathcal{V}^\rho_{\tau_1,\tau_1+\tau}\mathcal{Z}_X\mathcal{G}_{\tau_1}[v_0])(x)\right|^2\\
\le&\int\int\prod \mathbb{E}|G_{\tau_1+\tau}(x-y_i) - G_{\tau_1}(X-y_i)||v_0(y_i)|dy_1dy_2\\
+&\int_0^\tau\int \mathbb{E}\left|\sigma(\mathcal{V}_{\tau_1,\tau_1+r}\mathcal{G}_{\tau_1}[v_0]) - \sigma(\mathcal{V}_{\tau_1,\tau_1+r}\mathcal{Z}_X\mathcal{G}_{\tau_1}[v_0])\right|^2 G^2_{\tau+\rho-r}(x-y)dydr\\
=:& I_1 + I_2
\end{aligned}$$

Then we have $I_1 \le \dfrac{\tau + |X-x|^2/2}{\tau_1}\|v_0\|^2$ and for I_2, we have

$$I_2 \le \frac{\lambda^2}{4\pi}\int_0^\tau\int \frac{G_{(\tau+\rho-r)/2}(x-y)f(r,y)}{\tau+\rho-r}dydr$$

where

$$f(r,y) := \mathbb{E}\left|\mathcal{V}_{\tau_1,\tau_1+r}\mathcal{G}_{\tau_1}[v_0] - \mathcal{V}_{\tau_1,\tau_1+r}\mathcal{Z}_X\mathcal{G}_{\tau_1}[v_0]\right|^2$$

so for all $\tau \in [\tau_1, \tau_2]$, we have

$$f(\tau,x) \le \frac{\tau_2 - \tau_1 + |X-x|^2/2}{\tau_1}\|v_0\|_2^2 + \frac{\lambda^2}{4\pi}\int_0^\tau\int \frac{G_{(\tau+\rho-r)/2}(x-y)f(r,y)}{\tau+\rho-r}dydr$$

Assume $f(s,x) \le a(s) + b(s)|X-x|^2$, then we have

$$\begin{aligned}
f(s,x) &\le A + B|X-x|^2 + C\int_0^s\int \frac{G_{(\tau+\rho-r)/2}(x-y)(a(r)+b(r)|X-x|^2)}{s+\rho-r}dydr\\
&= A + B|X-x|^2 + C\int_0^s\left(\frac{a(r)+b(r)|X-x|^2}{s+\rho-r} + b(r)\right)dr
\end{aligned}$$

where $A = \dfrac{\tau_2-\tau_1}{\tau_1}\|v_0\|_2^2, B = \dfrac{1}{2\tau_1}\|v_0\|_2^2, C = \dfrac{\lambda^2}{4\pi}$.So for the optimal a, b, we may have

$$\begin{aligned}
a(s) &\le A + C\int_0^s \frac{a(r)}{s+\rho-r}dr + C\int_0^s b(r)dr\\
b(s) &\le B + C\int_0^s \frac{b(r)}{s+\rho-r}dr
\end{aligned}$$

Choose $\sup_{s\in[0,\tau_2-\tau_1]} b(s)$ and $\sup_{s\in[0,\tau_2-\tau_1]} a(s)$, we will have

$$\sup_{s\in[0,\tau_2-\tau_1]} b(s) \le B + C \sup_{s\in[0,\tau_2-\tau_1]} b(s)L(s/\rho)$$
$$\sup_{s\in[0,\tau_2-\tau_1]} a(s) \le A + sC \sup_{s\in[0,\tau_2,\tau_1]} b(s) + C \sup_{s\in[0,\tau_2-\tau_1]} a(s)L(s/\rho)$$

and hence $b(s) \le \dfrac{B}{1-CL((\tau_2-\tau_1)/\rho)}, a(s) \le \dfrac{A + s\dfrac{CB}{1-CL((\tau_2-\tau_1)/\rho)}}{1-CL((\tau_2-\tau_1)/\rho)}$. Let

$$\theta = \frac{\lambda^2}{4\pi} L_\rho(\tau_2-\tau_1)$$

we have

$$b(s) \le \frac{1}{2\tau_1(1-\theta)}\|v_0\|_2^2, \quad a(s) \le \frac{\tau_2-\tau_1}{\tau_1}\left(\frac{1}{1-\theta} + \frac{\lambda^2}{8\pi(1-\theta)^2}\right)$$

□

Proposition 6.3.3. Suppose $\lambda \in (0,\infty)$, $\mathrm{Lip}(\sigma) \le \lambda, \rho > 0$. Let $B \subset \mathbb{R}^2$ and let H be a half-plane in R^2. Let $\tau_0 < \tau_1 < \tau_0 + a$. Then for all $x \in H^c$ and $v_0 \in \mathscr{H}^2_{\tau_0}$, we have

$$\lambda^2 \mathbb{E}|(\mathring{\mathcal{V}}^B_{\tau_0,\tau_1} v_{\tau_0} - \mathring{\mathcal{V}}^{B/H}_{\tau_0,\tau_1} v_{\tau_0})(x)|^2 \le \frac{\lambda^2 S_\rho(a)}{4\pi - \lambda^2 L(a/\rho)} \Xi_{2;t}[v_0]^2.$$

Proof. For $t \in [0,\tau_1]$, we have

$$\begin{aligned}(\mathring{\mathcal{V}}^B_t v_0 - \mathring{\mathcal{V}}^{B/H}_t v_0)(x) = &\int_0^t \int_{B\cap H} G_{t+\rho-r}(x-y)\sigma(\mathring{\mathcal{V}}^B_r v_0(y))dW_r(y) \\ &+ \int_0^t \int_{B/H} G_{t+\rho-r}(x-y)[\sigma(\mathring{\mathcal{V}}^B_r v_0(y)) - \sigma(\mathring{\mathcal{V}}^{B/H}_r v_0(y))]dW_r(y)\end{aligned}$$

Then take the second moment

$$\begin{aligned}\mathbb{E}|(\mathring{\mathcal{V}}^B_t v_0 - \mathring{\mathcal{V}}^{B/H}_t v_0)(x)|^2 \le &\frac{\Xi_{2;t}[v_0]^2}{4\pi}\int_0^t \frac{G_{(t+\rho-r)/2}\chi_H(x)}{t+\rho-r}dr \\ &+ \frac{\lambda^2}{4\pi}\int_0^t \frac{\int G_{(t+\rho-r)/2}\mathbb{E}|(\mathring{\mathcal{V}}^B_r v_0 - \mathring{\mathcal{V}}^{B/H}_r v_0)(x)|^2 dy}{t+\rho-r}dr\end{aligned}$$

and we get

$$f_r(x) \le \mathcal{G}_{(t+\rho)/2}[\chi_H](x)\frac{\lambda^2 S_\rho(t)}{4\pi}\Xi_{2;t}[v_0]^2 + \frac{\lambda^2}{4\pi}\int_0^t \frac{\mathcal{G}_{(t+\rho-r)/2}[f_r](x)}{t+\rho-r}dr$$

where

$$f_t(x) := \lambda^2 \mathbb{E}|(\mathring{\mathcal{V}}^B_t v_0 - \mathring{\mathcal{V}}^{B/H}_t v_0)(x)|^2$$

Let

$$F(t) := \sup_{0\le r\le t} \|f_r\|_{L^\infty(\mathbb{R}^2)}.$$

Since the heat semigroup is an L^∞-contraction,

$$\|\mathcal{G}_s\,[g]\,\|_{L^\infty(\mathbb{R}^2)} \le \|g\|_{L^\infty(\mathbb{R}^2)},$$

and since $0 \le \chi_H \le 1$, we have

$$\|\mathcal{G}_{(s+\rho)/2}\,[\chi_H]\,\|_{L^\infty} \le 1.$$

Therefore, for every $0 \le s \le t$,

$$\begin{aligned}
\|f_s\|_{L^\infty} &\le \frac{\lambda^2 S_\rho(s)}{4\pi}\Xi_{2;t}[v_0]^2 + \frac{\lambda^2}{4\pi}\int_0^s \frac{\|\mathcal{G}_{(s+\rho-r)/2}\,[f_r]\,\|_{L^\infty}}{s+\rho-r}\,dr \\
&\le \frac{\lambda^2 S_\rho(t)}{4\pi}\Xi_{2;t}[v_0]^2 + \frac{\lambda^2}{4\pi}F(t)\int_0^s \frac{dr}{s+\rho-r}.\|f_s\|_{L^\infty} \\
&\le \frac{\lambda^2 S_\rho(t)}{4\pi}\Xi_{2;t}[v_0]^2 + \frac{\lambda^2}{4\pi}\log\left(1+\frac{t}{\rho}\right)F(t).
\end{aligned}$$

Taking the supremum over $0 \le s \le t$ yields

$$F(t) \le \frac{\lambda^2 S_\rho(t)}{4\pi}\Xi_{2;t}[v_0]^2 + \frac{\lambda^2}{4\pi}\log\left(1+\frac{t}{\rho}\right)F(t).$$

Therefore,

$$F(t) \le \frac{\lambda^2 S_\rho(t)}{4\pi\left(1-\dfrac{\lambda^2}{4\pi}L(\dfrac{t}{\rho})\right)}\Xi_{2;t}[v_0]^2.$$

□

Proposition 6.3.4. Suppose $\lambda \in (0,\infty)$, $\mathrm{Lip}(\sigma) \le \lambda, \rho > 0$. Let $R \subset \mathbb{R}^2$ to be a rectangle in R^2. Let $\tau_0 < \tau_1 < \tau_0 + a$. Then for all $x \in R$ and $v_0 \in \mathscr{H}^2_{\tau_0}$, we have

$$\lambda^2\mathbb{E}|(\mathcal{V}^\rho_{\tau_0,\tau_1}v_{\tau_0} - \mathcal{V}^{\rho,R}_{\tau_0,\tau_1}v_{\tau_0})(x)|^2 \le \frac{C\lambda^2 S_\rho(a)}{4\pi - \lambda^2 L(a/\rho)}\Xi_{2;t}[v_0]^2.$$

for some fixed constant C.

Definiton 6.3.1. Recall that

$$J_{\sigma,\rho}(q,a) = \left[\mathbb{E}\sigma^2(\mathcal{V}^{\sigma,\rho})\right]$$

$0 +$

Proposition 6.3.5. Suppose $\lambda \in (0,\infty)$, $\mathrm{Lip}(\sigma) \le \lambda, \rho > 0$. Then

6.4 The approxiamte root decoupling function

6.5 Approximation in the first short time stage

Let us recall the Theorem 5.14 from [**?**].

Theorem 6.5.1. Fix $M, \beta, \lambda \in (0,\infty), l \in (2,4], \overline{\beta} > \eta(\beta,l)$ where

$$\eta(\beta,l) = [2^{-2/l}l(l-1)(\beta^2+1-2/l)]^{1/2}$$

and $\overline{q} \in (0, \lambda^{-2})$. There is a constant $C = C(M, \beta, \lambda, l, \overline{\beta}, \overline{q}) \in (0, \infty)$ such that following holds. Suppose that $\rho \in (0, C^{-1})$. Let $\sigma \in \Sigma(M, \beta)$ satisfy $\mathrm{Lip}(\sigma) \leq \lambda$ and $(v_t)_{t \geq 0}$ solves

$$v_t^\rho = \mathcal{G}_{t-s}\left[v_s^\rho\right](x) + \gamma_\rho \int_s^t \mathcal{G}_{t+\rho-r}\left[\sigma(v_r^\rho) dW_r\right](x)$$

with initial condition $v_0 \in \mathscr{H}_0^l$, where $\mathscr{H}_t^l$ is the spaces of random fields z on $\mathbb{R}^2$ that are $\mathcal{F}_t$-measurable and satisfy

$$\|z\|_l := \sup_{x \in \mathbb{R}^2} (\mathbb{E}|z(x)|^l)^{1/l} < \infty$$

Let $\xi \in [0, \overline{T}_\rho(\overline{q}^{-1/2})]$, where $S_\rho := L(\cdot/\rho)/L(1/\rho), T_\rho := S_\rho^{-1} = \rho[(1 + 1/\rho)^\cdot - 1]$ and $\overline{T}_\rho := T_\rho((\cdot)^{-2} - 2L(1/\rho)^{-1})$,

$$t \in [([1 + L(1/\rho)]\kappa_{l,\rho}^{-1} + \kappa_{l,\rho})\xi^2, T_\rho(\overline{\beta}^{-2})],$$

where

$$\kappa_{l,\rho} = L(1/\rho)^{-1/(1-2/l)},$$

$x, z \in \mathbb{R}^2$ and $\iota_1, \iota_2 \in [0, 1]$. Then we have

$$\mathbb{E}|(S_{\xi,z}[\sigma^2 \circ v_t](x))^{1/2} - J_{\sigma,\rho}(S_\rho(L(1/\rho)^{\iota_1}\xi^2), \mathcal{G}_{L(1/\rho)^{\iota_2}\xi^2}\left[v_t\right](x))|_F^2 \leq C\langle \|v_0\|_l \rangle^2 \frac{L(L(1/\rho))}{L(1/\rho)}$$

where

$$J_{\sigma,\rho}(q, a) = [\mathbb{E}\sigma^2(\mathcal{V}_{T_\rho(q)}^{\sigma,\rho} a(x))]^{1/2}, \quad S_{\xi,z} f(x) := \frac{1}{\xi^2} \int_{\mathrm{B}_{\xi, k_{\xi,z}(x)+z}} f(y) dy$$

where $\mathrm{B}_{\xi,k} := \xi \cdot (k + [0,1)^2)$ for $k \in \mathbb{R}^2$ and $k_{\xi,z}(x)$ is the unique $k \in \mathbb{Z}^2$ such that $x \in \mathrm{B}_{\xi,k+z}$. And $(\mathcal{V}_{s,t}^{\sigma,\rho})_{-\infty < s < t < \infty}$ such that if $v_s \in \mathscr{H}_s^l$, then $v_t^\rho = \mathcal{V}_{s,t}^{\sigma,\rho} v_s$ is a solution of the integral equation above.

From the theorem, if we would like to apply the approximation to (**??**), then assume $\lambda := \mathrm{Lip}(\sigma)$ and we know

$$\mathrm{Lip}(\sigma_\rho) = \gamma_\rho^{-1}\lambda$$

and according to the proof of the Theorem 5.14 from [**?**], we may choose $\overline{q}$ such that

$$\overline{q} < (\alpha \mathrm{Lip}(\sigma_\rho))^{-2} - 2L(1/\rho)^{-1} = (4\pi/(\alpha\lambda)^2 - 2)L(1/\rho)^{-1}$$

where $\alpha \geq 1$, which means that only when $\lambda < \sqrt{2\pi}$, we may choose $\overline{q}$ to apply the proposition above. Then we may rewrite the theorem as

Theorem 6.5.2. For fixed σ with $\lambda := \mathrm{Lip}(\sigma) < \sqrt{\pi}$, where denote $p = \sqrt{2\pi}/\lambda$ and $M, \beta \in (0, \infty), l \in (2, 4], \overline{\beta} > \eta(\beta, l), \overline{q}_\rho := p^2 L(1/\rho)^{-1}$, there is a constant $C = C(p, \sigma, l, \beta, \overline{\beta})$ such that when $L(1/\rho) \geq 4\pi/\lambda^2 - 4$ and sufficiently small, the following holds. For $(v_t)_{t \geq 0}$ solves the integral equation

$$v_t^\rho = \mathcal{G}_{t-s}\left[v_s^\rho\right](x) + \int_s^t \mathcal{G}_{t+\rho-r}\left[\sigma(v_r^\rho) dW_r\right](x)$$

with initial condition $v_0 \in \mathscr{H}_0^l$. Let $\xi \in [0, \overline{T}_\rho(\overline{q}_\rho^{-1/2})]$

$$t \in [([1 + L(1/\rho)]\kappa_{l,\rho}^{-1} + \kappa_{l,\rho})\xi^2, T_\rho(\overline{\beta}^{-2})],$$

$x, z \in \mathbb{R}^2$ and $\iota_1, \iota_2 \in [0, 1]$. Then we have

$$\begin{aligned}&\mathbb{E}|(S_{\xi,z}[\sigma_\rho^2 \circ v_t](x))^{1/2} - J_{\sigma_\rho,\rho}(S_\rho(L(1/\rho)^{\iota_1}\xi^2), \mathcal{G}_{L(1/\rho)^{\iota_2}\xi^2}[v_t](x))|^2 \\ \leq & C(p, \sigma, l, \beta, \overline{\beta})\sqrt{L(L(1/\rho))}(L(1/\rho))^{l/4}\langle \|v_0\|_l \rangle^2\end{aligned}$$

and we are done.

Proof. Recall that

$$K_{\lambda,\rho,t} := (1 - \lambda^2[S_\rho(t) + 2L(1/\rho)^{-1}])^{-1} \quad \text{for } t \in [0, \overline{T}_\rho(\lambda)]$$

and according to the proof for the Theorem 5.14. from [**?**], the expectation we are approximating can be bounded by $E_1 + E_2 + E_3$.

For E_1, we have

$$E_1 \leq CK_{\gamma_\rho^{-1}\lambda,\rho,\xi^2}\Xi_{\tau_0,\tau_1}^\rho[v_{\tau_0}]^2 \left[L_\rho(\kappa_{l,\rho}^{-2}) + \kappa_{l,\rho}[(1 + \gamma_\rho^{-2}\lambda^2\delta)\kappa_{l,\rho} + 1] + \frac{K_{\gamma_\rho^{-1}\lambda,\rho,\xi^2}}{L(1/\rho)}\right]$$

for some fixed constant C and $\delta = K_{\gamma_\rho^{-1}\lambda,\rho,\kappa_{l,\rho}\xi^2}L(1/\rho)^{-1}$. And notice $K_{\lambda,\rho,t}$ is increasing related to t and $\xi \in [0, 1]$ when ρ is large enough, so

$$K_{\gamma_\rho^{-1}\lambda,\rho,\xi^2} \leq K_{\gamma^{-1}\lambda,\rho,\overline{q}_\rho} \leq p$$

For the rest terms, we know

$$\begin{aligned}L_\rho(\kappa_{l,\rho})^{-2} &\leq \frac{3}{1 - 2/l}\frac{L(L(1/\rho))}{L(1/\rho)} \\ \kappa_{l,\rho}[(1 + \gamma_\rho^{-2}\lambda^2\delta)\kappa_{l,\rho} + 1] &\leq \frac{2}{L(1/\rho)^{1/(1-2/l)}} \\ \frac{K_{\gamma_\rho^{-1}}}{L(1/\rho)} &\leq \frac{p}{L(1/\rho)}\end{aligned}$$

and so for ρ small enough we have

$$E_1 \leq C\frac{4p}{1 - 2/l}\frac{L(L(1/\rho))}{L(1/\rho)}\Xi_{\tau_0,\tau_1}^\rho[v_{\tau_0}]^2$$

for some fixed constant C.

For E_2, there are two cases. If $\kappa_{l,\rho}\xi^2 \geq \rho$ according to [**?**], we have

$$E_2 \leq C\Xi_{l;\kappa_{l,\rho}\xi^2}^\rho[\mathcal{G}_{\tau_1-\tau_0}[v_{\tau_0}(X)]^2]\frac{L(L(1/\rho))}{L(1/\rho)}$$

and for the other case, we have

$$E_2 \leq 4\Xi_{l;\kappa_{l,\rho}\xi^2}^\rho[\mathcal{G}_{\tau_1-\tau_0}[v_{\tau_0}(X)]^2]\gamma_\rho^2\lambda^{-2}\frac{L(2)}{L(1/\rho)}$$

Therefore, we have

$$E_2 \le C\Xi^{\rho}_{l;\kappa_{l,\rho}\xi^2}[\mathcal{G}_{\tau_1-\tau_0}\left[v_{\tau_0}(X)\right]^2]$$

where C depends only on l and λ where $X := (X_i(x))_{i=1}^N$ defined as in [**?**].

For E_3, we have

$$\begin{aligned} E_3 &\le \gamma_\rho^{-1}\lambda\sqrt{p}\left(\frac{C\sqrt{L(L(1/\rho))}}{\sqrt{L(1/\rho)}}\right)\left(\Xi^{\rho}_{2;L(1/\rho)\xi^2}[\mathcal{G}_{\kappa_{l,\rho}^{-1}\xi^2}\left[v_{\tau_0}(X)\right]]^2 + \Xi^{\rho}_{2;t}[v_0]\right) \\ &= C(p,\lambda,l)\sqrt{L(L(1/\rho))}\left(\Xi^{\rho}_{2;L(1/\rho)\xi^2}[\mathcal{G}_{\kappa_{l,\rho}^{-1}\xi^2}\left[v_{\tau_0}(X)\right]]^2 + \Xi^{\rho}_{2;t}[v_0]\right) \end{aligned}$$

For the term

$$\Xi^{\rho}_{\tau_0,\tau_1}[v_{\tau_0}]^2 + \Xi^{\rho}_{l;\kappa_{l,\rho}\xi^2}[\mathcal{G}_{\tau_1-\tau_0}\left[v_{\tau_0}(X)\right]^2] + \Xi^{\rho}_{2;L(1/\rho)\xi^2}[\mathcal{G}_{\kappa_{l,\rho}^{-1}\xi^2}\left[v_{\tau_0}(X)\right]]^2 + \Xi^{\rho}_{2;t}[v_0]$$

which should be $O(L(1/\rho)^{l/4})$ by the proposition 4.4 from [**?**],and finally we have

$$E_1 + E_2 + E_3 \le C(p,\sigma,l,\beta,\overline{\beta})\sqrt{L(L(1/\rho))}(L(1/\rho))^{l/4}\langle\|v_0\|_l\rangle^2$$

□

6.6 First Stage Approximation

Theorem 6.6.1.

6.7 Induction on scales

We will do the induction for extending the solution as Chapter 6 in [**?**], recall that

$$\beta_*(\sigma) := \inf\{\beta \in \mathbb{R}^+ : \sigma \in \Sigma(M,\beta) \text{ for some } M \in \mathbb{R}^+\}$$

where $\Sigma(M,\beta) := \{\sigma \in \mathrm{Lip}(\mathbb{R};\mathbb{R}) : \sigma(u)^2 \le M + \beta^2 u^2 \text{ for all } u \in \mathbb{R}\}$, and hence $\beta^*(\sigma_\rho) = +\infty$ comes up immediately. Then let us recall a proposition:

Proposition 6.7.1. For bounded continuous function σ, $Q_{\mathrm{FBSDE}}(\sigma) = \infty$.

Definiton 6.7.1. For a Borel set B $\subset \mathbb{R}$

Proposition 6.7.2.

Proof. Take $\overline{\beta} > \beta > 0$ and choose Q_2, Q_2' such that

$$Q_1 < Q_2 < Q_2' < [Q_1 + \mathrm{Lip}(J_\sigma(Q_1,\cdot))^{-2}] \wedge \beta^{-2}$$

Let $l \in (2,4]$ and l close to 2 such that $\overline{\beta} > \eta(\beta,l)$. □

A Fundamentals

A.1 Wiener Integral

Let T be a set and $X := \{X(t)\}_{t\in T}$ a T-indexed stochastic process. We recall that X is a Gaussian random field (process when $T \subset \mathbb{R}$) if $(X_{t_1}, \cdots, X_{t_m})$ is a Gaussian random vector for all $t_1, \cdots, t_m \in T$.

Definiton A.1.1. Let $\mathscr{B}(\mathbb{R}^m)$ denote the collection of all Borel-measurable subsets of $\mathbb{R}^m$ that have finite Lebesgue measure. White noise on $\mathbb{R}^m$ is a mean-zero set-indexed Gaussian random field $\xi(A)_{A\in\mathscr{B}(\mathbb{R}^m)}$ with covariance function

$$E[\xi(A_1)\xi(A_2)] := |A_1 \cap A_2| \quad \text{for all } A_1, A_2 \in \mathscr{L}(\mathbb{R}^m),$$

where $|\cdot|$ denotes the Lebesgue measure on $\mathbb{R}^m$ for every m.

Remark. Here is a general proof from Ash and Gardner.

Proof. For any complex numbers $c_1, \cdots, c_n$ and $A_1, \cdots, A_n \in \mathscr{B}(\mathbb{R}^m)$, we have

$$\sum_{1\le i,j\le n} c_i\bar{c_j}|A_i \cap A_j| = \sum_{1\le i,j\le n} c_i\bar{c_j}\int \chi(A_i)\chi(A_j) = \int \left|\sum_{i=1}^n c_i\chi(A_i)\right|^2 \ge 0$$

and we are done. □

Lemma A.1.1. $\{\xi(A)\}_{A\in\mathscr{B}(\mathbb{R}^m)}$ is an $L^2(\Omega)$-valued countably-additive measure on $\mathscr{B}(\mathbb{R}^m)$.

Definiton A.1.2 (Wiener Integral)**.** For $h \in \mathscr{S}(\mathbb{R}^m)$ the set of the simple functions, define

$$\int h d\xi := \sum_{i=1}^n c_i\xi(A_i)$$

where $h(x) = \sum_{i=1}^n c_i\chi(A_i)$ for some $A_1, \cdots, A_n \in \mathscr{B}(\mathbb{R}^m)$ and $c_1, \cdots, c_n \in \mathbb{R}$.

For $f \in L^2(\mathbb{R}^m)$, define

$$\int f d\xi = \lim_{n\to\infty}\int f_i d\xi$$

in $L^2(\Omega)$ for some $f_i \in \mathscr{S}(\mathbb{R}^m)$ and $f_i \to f$ in L^2.

Proof. □

Proposition A.1.2. The $L^2(\mathbb{R}^m)$-indexed Gaussian random field $\{\int h d\xi\}_{h\in L^2(\mathbb{R}^m)}$ has the following properties:

1. $\int(a_1h_1 + a_2h_2)d\xi = a_1\int h_1 d\xi + a_2\int h_2 d\xi$ a.s. for all $a_1, a_2 \in \mathbb{R}$ and $h_1, h_2 \in L^2(\mathbb{R}^m)$.
2. For all $h, h_1, h_2 \in L^2(\mathbb{R}^m)$,

$$\mathbb{E}\int h d\xi = 0, \mathbb{E}(\int h_1 d\xi \int h_2 d\xi) = \int h_1 h_2$$

Remark. We may define

$$\int_A h d\xi = \int h\chi(A) d\xi$$

and it is easy to prove the *Wiener isometry* by the properties above.

Definiton A.1.3. For every $f \in L^2(\mathbb{R}^m)$ and $x \in \mathbb{R}^m$, define

$$(f * \xi)(x) := \int f(x-y)\xi(dy)$$

Proposition A.1.3. For every $f \in L^2(\mathbb{R}^m)$, there exists a version of $f*\xi$ that is measurable jointly in (ω, x). Moreover, we have the following Young-type inequality: For every finite Borel measure μ on $\mathbb{R}^m$,

$$\mathbb{E}\left(\left|\int |(f*\xi)(x)|\mu(dx)\right|^2\right) \leq \mu(\mathbb{R}^m)^2 ||f||^2_{L^2(\mathbb{R}^m)}$$

Proof. □

A.2 Useful Approximations

Proposition A.2.1 (Bihari-LaSalle Inequality)**.**

Let u and f be non-negative continuous functions defined on the half-infinite ray $[0,\infty)$ and let w be a continuous non-decreasing function defined on $[0,\infty)$ and $w(u) > 0$ on $(0,\infty)$. For u we have

$$u(t) \leq \alpha + \int_0^t f(s)w(u(s))ds, \quad t \in [0,\infty),$$

where α is a non-negative constant, then

$$u(t) \leq G^{-1}\left(G(\alpha) + \int_0^t f(s)ds\right), \quad t \in [0,T]$$

where the function G is defined by

$$G(x) = \int_{x_0}^x \frac{dy}{w(y)}, \quad x \geq 0, x_0 \geq 0,$$

and G^{-1} is the inverse function of G and T is chosen so that

$$G(\alpha) + \int_0^t f(s)ds \in \mathrm{Dom}(G^{-1}), \quad \forall t \in [0,T]$$

Proof. Notice $G(x) = \int_{x_0}^x \frac{dy}{w(y)}$ is increasing and

$$\mathrm{Dom}(G^{-1}) = \left[G(0) = -\int_0^{x_0} \frac{dy}{w(y)}, G(0) + \int_0^\infty \frac{dy}{w(y)}\right]$$

then since $u \geq 0$, then we have

$$G(u(t)) \leq G\left(\alpha + \int_0^t f(s)\omega(u(s))ds\right) := H(t)$$

and we have

$$H'(t) = \frac{f(t)\omega(u(t))}{\omega(\alpha + \int_0^t f(s)\omega(u(s))ds)} \leq f(t)$$

so we have

$$H(t) \le G(\alpha) + \int_0^t f(s)ds.$$

Therefore, we may have the requested inequality for all $0 \le t \le T$ where

$$T := \sup_t \left\{ \int_0^t f(s)ds \le G(0) - G(\alpha) + \int_0^\infty \frac{dy}{\omega(y)} \right\}$$

□

Theorem A.2.2 (Poincare Inequality for Periodic Initial Condition Parabolic Equation)**.**
Let $\mathbb{T}_L = \mathbb{R}/L\mathbb{Z}$. If $f \in H^1(\mathbb{T}_L)$ and $\int_{\mathbb{T}_L} f(x)\,dx = 0$, then

$$\int_{\mathbb{T}_L} f(x)^2\,dx \le \left(\frac{L}{2\pi}\right)^2 \int_{\mathbb{T}_L} |\partial_x f(x)|^2\,dx$$

Proof. Since f is L-periodic, we may write its Fourier series as

$$f(x) = \sum_{k\in\mathbb{Z}} \widehat{f}_k e^{2\pi ikx/L},$$

where

$$\widehat{f}_k = \frac{1}{L}\int_0^L f(x)e^{-2\pi ikx/L}\,dx, \quad \widehat{f}_0 = 0$$

and hence

$$f(x) = \sum_{k\in\mathbb{Z}\setminus\{0\}} \widehat{f}_k e^{2\pi ikx/L}$$

Since $\partial_x f$ is L^2 on $\mathbb{T}_L$, we have

$$\partial_x f(x) = \sum_{k\in\mathbb{Z}\setminus\{0\}} \frac{2\pi ik}{L}\widehat{f}_k e^{2\pi ikx/L}.$$

by integration by parts. Then Parseval's identity gives

$$\int_0^L |f(x)|^2\,dx = L\sum_{k\neq 0} |\widehat{f}_k|^2, \quad \int_0^L |\partial_x f(x)|^2\,dx = L\sum_{k\neq 0} \left|\frac{2\pi k}{L}\right|^2 |\widehat{f}_k|^2.$$

It follows that

$$\begin{aligned}\int_0^L |\partial_x f(x)|^2\,dx &= L\sum_{k\neq 0} \left|\frac{2\pi k}{L}\right|^2 |\widehat{f}_k|^2 \ge L\sum_{k\neq 0}\left(\frac{2\pi}{L}\right)^2 |\widehat{f}_k|^2 \\ &= \left(\frac{2\pi}{L}\right)^2 L\sum_{k\neq 0}|\widehat{f}_k|^2 = \left(\frac{2\pi}{L}\right)^2 \int_0^L |f(x)|^2\,dx.\end{aligned}$$

Thus

$$\int_0^L |f(x)|^2\,dx \le \left(\frac{L}{2\pi}\right)^2 \int_0^L |\partial_x f(x)|^2\,dx.$$

□

Proposition A.2.3. For (t,x) in $\mathbb{R}^+ \times \mathbb{T}_L, L > 0$, consider the parabolic equation

$$\partial_t u - \partial_x(a_1(t,x)\partial_x u + a_2(t,x)u) - b(t,x)\partial_x u - c(t,x)u = 0$$

with the coefficients a_1, a_2, b, c are bounded measurable functions in S and that there exists a constant $\nu > 0$ such that $a_1 \geq \nu^{-1}$ almost everywhere in S. Assume u is a solution of the parabolic equation above, then there is a constant C such that

$$\|u(t,\cdot)\|_{L^\infty(\mathbb{T}_L)} \leq C\|u(t-1,\cdot)\|_{L^2(\mathbb{T}_L)}$$

for any $t \geq 1$.

Proof. By the Theorem.1. from [**?**], we apply it to the strip $(t-1, t+1) \times \mathbb{R}$ with initial condition $\tilde{u}(t,\tilde{x}), \tilde{x} \in \mathbb{R}$, which is a periodic expansion of $u(t,x), x \in \mathbb{T}_L$ where $[\tilde{x}]_{\mathbb{T}_L} = x$. The fundamental solution $g(x,t;\xi,\tau)$ on this strip admits the inequality

$$0 \leq g(x,t;\xi,\tau) \leq K\gamma(x-\xi, t-\tau)$$

for some constant K and a heat kernel γ with coeffients only related to ν and the bound of a_1, a_2, b, c, so K and γ are uniform to t. Then we know

$$\tilde{u}(t,x) = \int_{\mathbb{R}} g(x,1;y,0)\tilde{u}(t-1,y)dy$$

and hence

$$\begin{aligned}
|\tilde{u}(t,x)| &\leq K\int_{\mathbb{R}} \gamma(x-y,1)|\tilde{u}(t-1,y)|dy \\
&= K\sum_{k\in\mathbb{Z}} \int_{[(k-1)L,kL]} \gamma(x-y,1)|\tilde{u}(t-1,y)|dy \\
\text{(by Holder Inequality)} &\leq K\sum_{k\in\mathbb{Z}} \|\gamma(x-\cdot,1)\|_{L^2([(k-1)L,kL])}\|u(t-1)\|_{L^2(\mathbb{T}_L)} \\
\text{(by Triangle Inequality)} &\leq K\|\gamma(\cdot,1)\|_{L^2(\mathbb{R})}\|u(t-1)\|_{L^2(\mathbb{T}_L)}
\end{aligned}$$

Combing K and $\|\gamma(\cdot,1)\|_{L^2}$, which both are uniform to t, we obtain the desired inequality from the proposition. □

A.3 Matrix Analysis

Theorem A.3.1 (Spectral Theorem for Real Symmetric Matrices)**.**
Let $A \in \mathbb{R}^{n\times n}$ be a real symmetric matrix, that is, $A^T = A$. Then:

1. All eigenvalues of A are real.
2. Eigenspaces corresponding to distinct eigenvalues are orthogonal.
3. The space $\mathbb{R}^n$ decomposes as an orthogonal direct sum of eigenspaces:

$$\mathbb{R}^n = \bigoplus_{\lambda\in\sigma(A)} \ker(A - \lambda I).$$

4. Equivalently, there exists an orthogonal matrix Q and a real diagonal matrix Λ such that

$$A = Q\Lambda Q^T.$$

A.4 Kolmogorov's Theorems

Definiton A.4.1. For any $w \in \mathbb{R}^m$, then

$$\rho(w) := \sum_{i=1}^{m} |w_j|^{\alpha_j}$$

where $0 < \alpha_1, \cdots, \alpha_m \leq 1$ are fixed.

For a bounded and closed subset T of $\mathbb{R}^m$, define

$$\Lambda_T(a) := \int_T dt \int_T ds[\rho(t-s)]^{-a} \quad \text{for all } a > 0.$$

Remark. $\Lambda_T(a) < \infty$ if and only if $a < H := \sum_{j=1}^{m} \alpha_j^{-1}$.

Theorem A.4.1 (Kolmogorov's Continuity Theorem)**.**

Let $\{X_t\}_{t\in T}$ be a stochastic process, where $T \subset \mathbb{R}^m$ is measurable and bounded, and suppose that there exists finite real numbers $C > 0$ and $k > H$, such that

$$\|X_t - X_s\|_k \leq C\rho(t-s) \quad \text{for any } s, t \in T$$

Then X has a modification $\overline{X}$ that is Holder continuous a.s. Moreover, $\overline{X}$ satisfies the following: For all $q \in (0, 1-(H/k))$ and $\delta \in (q, 1-(H/k))$ there exists a finite constant D, that does not depend on that numerical value of k, such that

$$\mathbb{E}\left(\sup_{s,t\in T, s\neq t} \left|\frac{\overline{X}_t - \overline{X}_s}{[\rho(t-s)]^q}\right|^k\right) \leq (\delta - q)^{-1}\left(\frac{DC}{\delta}\right)^k k^{2H}\Lambda_T(2H - k + k\delta)$$